A First Course in Abstract Algebra

Part 1: Groups and Subgroups

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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1 Introduction and Examples

Exercise 1.1: A cube of i

$i^3 = i^2 i = -i$, so the answer is $0 - i$.

Exercise 1.2: A fourth power of i

$i^4 = (i^2)^2 = (-1)^2 = 1$, or $1 + 0i$.

Exercise 1.3: A high power of i

Since $23 = 4 \cdot 5 + 3$ and $i^4 = 1$, $i^{23} = (i^4)^5 i^3 = -i$.

Exercise 1.4: A power of minus i

$(-i)^{35} = (-1)^{35} i^{35} = -i^3 = i$, because $35 \equiv 3 \pmod 4$.

Exercise 1.5: Multiplying two binomials

Distribute and use $i^2 = -1$: $(4 - i)(5 + 3i) = 20 + 12i - 5i - 3i^2 = 23 + 7i$.

Exercise 1.6: Another binomial product

$(8 + 2i)(3 - i) = 24 - 8i + 6i - 2i^2 = 26 - 2i$. The real contribution from $-2i^2$ is $+2$.

Exercise 1.7: A product plus a complex number

$(2 - 3i)(4 + i) = 8 + 2i - 12i - 3i^2 = 11 - 10i$. Adding $6 - 5i$ yields $17 - 15i$.

Exercise 1.8: Cubing one plus i

First $(1 + i)^2 = 1 + 2i + i^2 = 2i$. Thus $(1 + i)^3 = 2i(1 + i) = -2 + 2i$.

Exercise 1.9: A fifth power by the binomial theorem

Expand $(1 - i)^5 = 1 - 5i + 10i^2 - 10i^3 + 5i^4 - i^5$. Replacing $i^2 = -1, i^3 = -i, i^4 = 1, i^5 = i$ gives $(1 - 10 + 5) + (-5 + 10 - 1)i = -4 + 4i$.

Exercise 1.10: The modulus of three minus four i

For $a + bi$ the modulus is $\sqrt{a^2 + b^2}$. Hence $|3 - 4i| = \sqrt{9 + 16} = 5$.

Exercise 1.11: The modulus of six plus four i

$|6 + 4i| = \sqrt{36 + 16} = \sqrt{52} = 2\sqrt{13}$. The modulus is nonnegative.

Exercise 1.12: Polar form: Three minus four i

Compute $r = |3 - 4i| = 5$ and divide the real and imaginary coordinates by r. This gives

$$3 - 4i = 5\left(3/5 + (-4/5)\, i\right).$$

The two displayed normalized coordinates have squared sum 1, so the parenthesized complex number lies on the unit circle. This is the requested form $r(p + qi)$.

Exercise 1.13: Polar form: Minus one plus i

Compute $r = |-1 + i| = \sqrt{2}$ and divide the real and imaginary coordinates by r. This gives

$$-1 + i = \sqrt{2}\left(-1/\sqrt{2} + \left(1/\sqrt{2}\right) i\right).$$

The two displayed normalized coordinates have squared sum 1, so the parenthesized complex number lies on the unit circle. This is the requested form $r(p + qi)$.

Exercise 1.14: Polar form: Twelve plus five i

Compute $r = |12 + 5i| = 13$ and divide the real and imaginary coordinates by r. This gives

$$12 + 5i = 13\left(12/13 + (5/13)\, i\right).$$

The two displayed normalized coordinates have squared sum 1, so the parenthesized complex number lies on the unit circle. This is the requested form $r(p + qi)$.

Exercise 1.15: Polar form: Minus three plus five i

Compute $r = |-3 + 5i| = \sqrt{34}$ and divide the real and imaginary coordinates by r. This gives

$$-3 + 5i = \sqrt{34}\left(-3/\sqrt{34} + \left(5/\sqrt{34}\right) i\right).$$

The two displayed normalized coordinates have squared sum 1, so the parenthesized complex number lies on the unit circle. This is the requested form $r(p + qi)$.

Exercise 1.16: Fourth roots of one

Write $z = re^{i\theta}$. Then $r^4 = 1$, so $r = 1$, and $4\theta = 2\pi k$. Distinct roots occur for $k = 0, 1, 2, 3$:

$$z = e^{i\pi k/2} \in \{1, i, -1, -i\}.$$

Larger integers repeat these roots; a fourth-degree polynomial has no more than four distinct roots.

Exercise 1.17: Fourth roots of minus one

Since $-1 = e^{i(\pi+2\pi k)}$, the roots are

$$z = e^{i(\pi/4+k\pi/2)}, \quad k = 0, 1, 2, 3.$$

In Cartesian form these are $(1+i)/\sqrt{2}$, $(-1+i)/\sqrt{2}$, $(-1-i)/\sqrt{2}$, and $(1-i)/\sqrt{2}$. Each has modulus 1 and fourth power -1.

Exercise 1.18: Cube roots of minus eight

Here $-8 = 8e^{i(\pi+2\pi k)}$. Thus $r = 2$ and $\theta = \pi/3 + 2\pi k/3$. For $k = 0, 1, 2$ the roots are $1 + i\sqrt{3}$, -2, and $1 - i\sqrt{3}$. Distinct arguments modulo 2π give all three solutions.

Exercise 1.19: Cube roots of minus twenty-seven i

Use $-27i = 27e^{i(-\pi/2+2\pi k)}$. The roots are $3e^{i(-\pi/6+2\pi k/3)}$ for $k = 0, 1, 2$, namely

$$\frac{3\sqrt{3}}{2} - \frac{3}{2}i, \qquad 3i, \qquad -\frac{3\sqrt{3}}{2} - \frac{3}{2}i.$$

Cubing triples each argument and cubes the modulus, giving $-27i$ in every case.

Exercise 1.20: Sixth roots of one

The six roots are $e^{i\pi k/3}$, $k = 0, \ldots, 5$. Explicitly:

$$1, \quad \tfrac{1}{2} + \tfrac{\sqrt{3}}{2}i, \quad -\tfrac{1}{2} + \tfrac{\sqrt{3}}{2}i, \quad -1, \quad -\tfrac{1}{2} - \tfrac{\sqrt{3}}{2}i, \quad \tfrac{1}{2} - \tfrac{\sqrt{3}}{2}i.$$

They follow from $r^6 = 1$ and $6\theta \in 2\pi\mathbb{Z}$.

Exercise 1.21: Sixth roots of minus sixty-four

Here $r^6 = 64$ gives $r = 2$ and $6\theta = \pi + 2\pi k$. Thus $z = 2e^{i(\pi/6+k\pi/3)}$, $k = 0, \ldots, 5$, or

$$\sqrt{3} + i, \quad 2i, \quad -\sqrt{3} + i, \quad -\sqrt{3} - i, \quad -2i, \quad \sqrt{3} - i.$$

These have six distinct arguments modulo 2π.

Exercise 1.22: Addition modulo seventeen

$10 + 16 = 26 = 17 + 9$, so $10 +_{17} 16 = 9$. In modular addition, choose the unique representative between zero inclusive and the modulus exclusive.

Exercise 1.23: Addition modulo ten

$8 + 6 = 14 = 10 + 4$, so $8 +_{10} 6 = 4$. In modular addition, choose the unique representative between zero inclusive and the modulus exclusive.

Exercise 1.24: Real modular addition

$20.5 + 19.3 = 39.8 = 25 + 14.8$, so the representative in $[0, 25)$ is 14.8. In modular addition, choose the unique representative between zero inclusive and the modulus exclusive.

Exercise 1.25: Addition modulo one

$1/2 + 7/8 = 11/8 = 1 + 3/8$, hence $1/2 +_1 7/8 = 3/8$. In modular addition, choose the unique representative between zero inclusive and the modulus exclusive.

Exercise 1.26: Angles modulo two pi

$3\pi/4 + 3\pi/2 = 9\pi/4 = 2\pi + \pi/4$, so the result is $\pi/4$. In modular addition, choose the unique representative between zero inclusive and the modulus exclusive.

Exercise 1.27: An irrational modulus

$2\sqrt{2} + 3\sqrt{2} = 5\sqrt{2}$, and $\sqrt{32} = 4\sqrt{2}$. Subtracting that modulus once leaves $\sqrt{2} \in [0, 4\sqrt{2})$, the answer. In modular addition, choose the unique representative between zero inclusive and the modulus exclusive.

Exercise 1.28: Inputs outside the underlying set

The operation $+_6$ on $\mathbb{R}_6 = [0, 6)$ accepts only ordered pairs of elements of that set. The number 8 is not an element of $\mathbb{R}_6$, so $5 +_6 8$ is not an operation on the stated domain. One could first replace 8 by 2, but that is an additional reduction, not the given expression as written.

Exercise 1.29: A linear equation modulo fifteen

The congruence is $x + 7 \equiv 3 \pmod{15}$, so $x \equiv -4 \equiv 11 \pmod{15}$. The only representative in $\mathbb{Z}_{15}$ is 11, and $11 +_{15} 7 = 3$.

Exercise 1.30: An angular equation

Subtract $3\pi/2$ from $3\pi/4$: $x \equiv -3\pi/4 \pmod{2\pi}$. Adding 2π gives $x = 5\pi/4 \in [0, 2\pi)$. Its sum with $3\pi/2$ is $11\pi/4$, whose remainder is $3\pi/4$.

Exercise 1.31: Twice a residue modulo seven

We need $2x \equiv 3 \pmod{7}$. Since $2 \cdot 4 \equiv 1$, multiplication by 4 gives $x \equiv 12 \equiv 5$. Checking, $5 +_7 5 = 3$. The inverse of 2 makes the answer unique.

Exercise 1.32: Three times a residue modulo seven

The equation is $3x \equiv 5 \pmod 7$. Since $3 \cdot 5 \equiv 1$, we get $x \equiv 25 \equiv 4$. Indeed $4 + 4 + 4 = 12 \equiv 5 \pmod 7$.

Exercise 1.33: Twice a residue modulo twelve

The equation $2x \equiv 2 \pmod{12}$ means $12 \mid 2(x-1)$, equivalently $6 \mid x - 1$. Representatives in $\mathbb{Z}_{12}$ are therefore $x = 1$ and $x = 7$. Both give remainder 2 when doubled. Dividing a congruence here also divides its modulus.

Exercise 1.34: Four times a residue modulo four

Every residue $x \in \{0, 1, 2, 3\}$ satisfies $4x \equiv 0 \pmod 4$. Hence all four residues are solutions; the coefficient has no multiplicative inverse modulo 4.

Exercise 1.35: An isomorphism from eighth roots

Multiplication of powers corresponds to addition, so the image of ζ^m must be $5m \pmod 8$. For the six requested exponents:

m	0	3	4	5	6	7
$5m \bmod 8$	0	7	4	1	6	3

This also gives $m = 2 \mapsto 2$, as required in the question. Since $\gcd(5, 8) = 1$, all eight images are distinct.

Exercise 1.36: An isomorphism from seventh roots

The image of ζ^m is $4m \pmod 7$ because powers become repeated sums. Thus

m	0	2	3	4	5	6
$4m \bmod 7$	0	1	5	2	6	3

Multiplication by 4 is a permutation of $\mathbb{Z}_7$, since 4 is relatively prime to 7.

Exercise 1.37: Why the proposed image is impossible

The element $\zeta = e^{i\pi/3}$ generates all six roots, so its image under an isomorphism must generate $\mathbb{Z}_6$. But 4 generates only $\{0, 4, 2\}$, since $3 \cdot 4 \equiv 0 \pmod 6$. Alternatively, $\zeta^3 = -1 \neq 1$ would map to 0, violating injectivity. Thus no isomorphism can have this image.

Exercise 1.38: Angle addition from Euler formula

Compute the product in two ways:

$$e^{ia}e^{ib} = e^{i(a+b)} = \cos(a + b) + i \sin(a + b),$$

whereas direct multiplication gives

$$(\cos a + i\sin a)(\cos b + i\sin b) = (\cos a\cos b - \sin a\sin b) + i(\sin a\cos b + \cos a\sin b).$$

Equality of real parts proves the cosine identity; equality of imaginary parts proves the sine identity. This argument explains why multiplying unit complex numbers adds their angles.

Exercise 1.39: Multiplying polar forms

Put $r_j = |z_j|$. Distributing the factors gives

$$z_1 z_2 = r_1 r_2\{(\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2) + i(\sin\theta_1\cos\theta_2 + \cos\theta_1\sin\theta_2)\}.$$

The two addition identities replace the braces by $\cos(\theta_1+\theta_2) + i\sin(\theta_1+\theta_2)$. Thus the moduli multiply and the angles add. If one number is zero, the same product formula is zero regardless of the chosen angle.

Exercise 1.40: The triple-angle identity

(a) Cubing Euler's formula gives $e^{3i\theta} = (\cos\theta + i\sin\theta)^3$. Its real part is $\cos^3\theta - 3\cos\theta\sin^2\theta$, since the terms with odd powers of i are imaginary and $i^2 = -1$. Hence $\cos 3\theta = \cos^3\theta - 3\cos\theta\sin^2\theta$.

(b) Substitute $\sin^2\theta = 1 - \cos^2\theta$ to obtain $\cos 3\theta = \cos^3\theta - 3\cos\theta(1-\cos^2\theta) = 4\cos^3\theta - 3\cos\theta$.

Exercise 1.41: Euler's formula from power series

Start with $e^z = \sum_{n=0}^{\infty} z^n/n!$ and substitute $z = i\theta$. Separate even and odd powers using $i^{2k} = (-1)^k$ and $i^{2k+1} = i(-1)^k$:

$$\begin{aligned} e^{i\theta} &= \sum_{k=0}^{\infty} \frac{(-1)^k\theta^{2k}}{(2k)!} + i\sum_{k=0}^{\infty}\frac{(-1)^k\theta^{2k+1}}{(2k+1)!} \\ &= \cos\theta + i\sin\theta. \end{aligned}$$

The two real series are exactly the given cosine and sine expansions. This is the requested formal derivation; their absolute convergence for every real θ also justifies separating the two subseries analytically.

2 Binary Operations

Exercise 2.1: Reading the operation table

Use the first factor's row and the second factor's column. Table 2.26 gives $b * d = e$ and $c * c = b$. For the nested expression, work from the innermost parentheses: $a * c = c$, then $c * e = a$, then $a * a = a$. Thus $[(a * c) * e] * a = a$.

Exercise 2.2: A test that happens to agree

The table gives $(a * b) * c = b * c = a$ and $a * (b * c) = a * a = a$. These values agree, but one successful triple does not establish associativity: that law must hold for every ordered triple. This computation alone therefore cannot decide whether the operation is associative.

Exercise 2.3: A failed associativity test

Since $b * d = e$, $(b * d) * c = e * c = a$. On the other hand $d * c = b$, so $b * (d * c) = b * b = c$. The elements a, c differ, so this one triple disproves associativity.

Exercise 2.4: Testing commutativity

The operation is not commutative: $a * e = d$ whereas $e * a = b$. A commutative table must be symmetric across its main diagonal; these two off-diagonal entries already violate that requirement.

Exercise 2.5: A symmetric completion

Reflect each known entry across the diagonal, leaving the given diagonal entries unchanged. This forces

$*$	a	b	c	d
a	a	b	c	d
b	b	d	a	c
c	c	a	d	b
d	d	c	b	a

For instance $c * b = a$ forces $b * c = a$, and $b * d = c$ forces $d * b = c$. Every resulting entry is in the specified set and the table is symmetric, as required.

Exercise 2.6: An associative completion

The known entry $c * b = d$ lets us compute every missing product: associativity requires $d * x = (c * b) * x = c * (b * x)$. For $x = a, b, c, d$, the given row of b is b, a, c, d; applying the given row of c gives d, c, c, d. Thus

$*$	a	b	c	d
a	a	b	c	d
b	b	a	c	d
c	c	d	c	d
d	d	c	c	d

The question permits assuming an associative completion exists. The preceding equations show it must be this one; direct checking of its 4^3 triples confirms it is associative.

Exercise 2.7: Subtraction on the integers

This is neither commutative nor associative. For commutativity, $1 * 2 = -1$ while $2 * 1 = 1$. For associativity, $(1 * 2) * 3 = -1 - 3 = -4$, whereas $1 * (2 * 3) = 1 - (-1) = 2$. Every output is still an integer, so it is a binary operation despite these failed laws.

Exercise 2.8: Multiplication followed by adding one

Commutativity holds because $ab + 1 = ba + 1$. However

$$(a * b) * c = (ab + 1)c + 1 = abc + c + 1, \qquad a * (b * c) = a(bc + 1) + 1 = abc + a + 1.$$

Taking $a = 0, b = 0, c = 1$ gives 2 versus 1, so associativity fails.

Exercise 2.9: A scaled product

Commutativity follows from $ab/2 = ba/2$. For every rational a, b, c,

$$(a * b) * c = ((ab)/2)c/2 = abc/4 = a((bc)/2)/2 = a * (b * c).$$

Thus the operation is both commutative and associative. Products and division by 2 keep all outputs rational.

Exercise 2.10: Two raised to a product

The operation is commutative since $ab = ba$. It is not associative: with $(a, b, c) = (1, 1, 3)$,

$$(1 * 1) * 3 = 2 * 3 = 2^6 = 64, \qquad 1 * (1 * 3) = 1 * 8 = 2^8 = 256.$$

All outputs are positive integers, so closure is not an issue.

Exercise 2.11: Exponentiation

This operation is not commutative: $2^3 = 8 \neq 9 = 3^2$. It is not associative either:

$$(2 * 2) * 3 = 4^3 = 64, \qquad 2 * (2 * 3) = 2^8 = 256.$$

The distinction is between $(a^b)^c = a^{bc}$ and a^{b^c}, not a failure of closure.

Exercise 2.12: Counting all binary operations

For a set of size n, there are n^2 ordered input pairs and independently n choices for each output. Thus there are n^{n^2} operations. For one, two, and three elements this gives respectively 1, $2^4 = 16$, and $3^9 = 19683$. The table is an arbitrary function from $S \times S$ into S.

Exercise 2.13: Counting commutative operations

Only the diagonal and one half of the off-diagonal entries can be chosen freely. There are $n + n(n-1)/2 = n(n+1)/2$ such positions, each with n possible outputs, so the answer is $n^{n(n+1)/2}$. It is $2^3 = 8$ for two elements and $3^6 = 729$ for three elements. The remaining entries are forced by symmetry.

Exercise 2.14: A precise commutativity definition

A binary operation $*$ on a set S is commutative if and only if $a * b = b * a$ *for all* $a, b \in S$. The universal quantifier and the underlying set must be specified: agreement for just one pair does not suffice.

Exercise 2.15: An acceptable associativity definition

The given definition is correct: $(b * c) * a = b * (c * a)$ for all $a, b, c \in S$ is the associative law with the dummy variables renamed. For the more usual arrangement, one may write $(x * y) * z = x * (y * z)$ for all $x, y, z \in S$. No correction of its mathematical content is needed.

Exercise 2.16: Closure of a subset

A subset $H \subseteq S$ is closed under $*$ if and only if $a * b \in H$ for all $a, b \in H$. The proposed version incorrectly quantifies over all of S. Closure only asks about two inputs already in H.

Exercise 2.17: A subtraction rule on positive integers

The numerical output $a - b$ is uniquely determined for every input, so Condition 1 holds. Condition 2, closure, fails: $1 * 1 = 0 \notin \mathbb{Z}^+$ (and $1 * 2 = -1$ is another failure). Thus this does not define a binary operation on $\mathbb{Z}^+$.

Exercise 2.18: An exponentiation rule on positive integers

For each positive integer pair a, b, the integer a^b exists uniquely and is positive. Both uniqueness/everywhere-definition (Condition 1) and closure (Condition 2) hold. Hence this is a binary operation on the given set.

Exercise 2.19: A subtraction rule on all real numbers

For any two real numbers a, b, the difference $a - b$ is one uniquely determined real number. Therefore Conditions 1 and 2 both hold; subtraction is a binary operation on $\mathbb{R}$. Associativity is not required in the definition of binary operation.

Exercise 2.20: The least integer above two inputs

For positive integers a, b, the specified integer is exactly $\max(a, b) + 1$. It exists, is unique, and is positive. Thus both conditions hold and the rule defines a binary operation.

Exercise 2.21: An ambiguous lower bound

The condition permits every integer $c \geq a + b + 5$. For example, when $a = b = 1$, both 7 and 8 are allowed. Thus Condition 1 fails because no unique output is specified. All allowed outputs are positive, so there is no violation of Condition 2. The rule is not a binary operation.

Exercise 2.22: The largest integer below a product

Since ab is an integer, the largest integer strictly below it is uniquely $ab - 1$. Thus Condition 1 holds. But $1 * 1 = 0$ is not positive, so Condition 2 fails and the rule is not a binary operation on $\mathbb{Z}^+$.

Exercise 2.23: A matrix-shaped subset

Write $M(a, b) = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$. For arbitrary real a, b, c, d,

$$M(a, b) + M(c, d) = M(a + c, b + d),$$

so **(a)** it is closed under addition. Also

$$M(a, b)M(c, d) = \begin{pmatrix} ac - bd & -(ad + bc) \\ ad + bc & ac - bd \end{pmatrix} = M(ac - bd, ad + bc),$$

so **(b)** it is closed under multiplication. Exhibiting the same form proves closure for all inputs.

Exercise 2.24: Ten statements about operations

(a) False: addition on $\mathbb{Z}$ has $1 + 1 \neq 1$.

(b) True: treat $b * c \in S$ as one element and apply commutativity to it and a.

(c) False: matrix multiplication is associative but generally noncommutative; specifically take $a = E_{12}$, $b = E_{21}$, $c = I_2$. Then $a(bc) = E_{11}$ but $(bc)a = E_{22}$.

(d) False: function composition, set union, and matrix multiplication are important operations on nonnumeric objects.

(e) False: the equation must hold for every pair. Subtraction has $0-0 = 0-0$ but is not commutative.

(f) True: on $\{s\}$ every product must be s, so both laws hold.

(g) True: exactly one output in particular means at least one.

(h) True: exactly one output in particular means at most one.

(i) True: this is the defining requirement, together with the specified codomain.

(j) False: multiple allowed outputs contradict being a function.

Exercise 2.25: Two operations on a nonnumeric set

Take S to be the set of all finite words in the alphabet $\{A, B\}$, including the empty word. Define $u * v$ to be concatenation and $u *' v = u$. Each ordered pair has exactly one output in S. They are different: for $u = A, v = B$, concatenation gives AB, while the second operation gives A. The alphabet and finiteness requirement make the set unambiguous.

Exercise 2.26: Rearranging four factors

Use only the stated laws, one step at a time:

$$\begin{aligned}(a * b) * (c * d) &= (c * d) * (a * b) && \text{(commutativity)}\\ &= ((c * d) * a) * b && \text{(associativity on } c * d, a, b)\\ &= ((d * c) * a) * b && \text{(commutativity on } c, d).\end{aligned}$$

Each parenthesized product is an element of S by closure, so applying the laws to such intermediate elements is legitimate.

Exercise 2.27: The singleton case

Let $S = \{s\}$. The only possible output of $s * s$ is s. Hence $s * s = s * s$ proves commutativity, and $(s * s) * s = s * s = s = s * (s * s)$ proves associativity. These exhaust all pairs and triples because the set has one element.

Exercise 2.28: Commutativity need not imply associativity

On $\{0, 1\}$ define $a * b = 1 - ab$ (the values are again 0 or 1). It is commutative because $ab = ba$. But $(0 * 0) * 1 = 1 * 1 = 0$, while $0 * (0 * 1) = 0 * 1 = 1$. Thus this commutative two-element operation is not associative.

Exercise 2.29: Associativity of function addition

For every real x, $((f + g) + h)(x) = (f(x) + g(x)) + h(x) = f(x) + (g(x) + h(x)) = (f + (g + h))(x)$. The middle equality is associativity of real addition. Since the two functions agree at every input, they are equal, so function addition is associative.

Exercise 2.30: Commutativity of function subtraction

False. Let $f(x) = 1$ and $g(x) = 0$ for every real x. Then $(f - g)(x) = 1$ while $(g - f)(x) = -1$, so the resulting functions differ. Constant functions are allowed elements of F.

Exercise 2.31: Associativity of function subtraction

False. Let f, g, h all be the constant function 1. Then $(f - g) - h$ is the constant -1, whereas $f - (g - h)$ is the constant 1. This counterexample uses functions with exactly the required domain and codomain.

Exercise 2.32: Commutativity of function multiplication

For every real x, $(fg)(x) = f(x)g(x) = g(x)f(x) = (gf)(x)$ by commutativity of real multiplication. Pointwise agreement proves $fg = gf$ as functions. Thus the statement is true.

Exercise 2.33: Associativity of function multiplication

For every x, $((fg)h)(x) = (f(x)g(x))h(x) = f(x)(g(x)h(x)) = (f(gh))(x)$. Associativity of real multiplication gives the central equality. Thus the two functions agree on all of $\mathbb{R}$, proving the assertion.

Exercise 2.34: Commutativity of composition

False. Set $f(x) = x + 1$ and $g(x) = 2x$. Then $(f \circ g)(x) = 2x + 1$ but $(g \circ f)(x) = 2x + 2$. The order of applying functions matters, even though composition is associative.

Exercise 2.35: An arbitrary distributive law

False: choose both operations to be ordinary addition on $\mathbb{Z}$. With $a = 1, b = c = 0$, the left side $a + (b + c)$ is 1, whereas $(a + b) + (a + c)$ is 2. Distributivity is an extra relation between operations; it does not follow from their merely being binary operations.

Exercise 2.36: Closure of elements commuting with everything

Let $a, b \in H$ and $x \in S$. Reassociate and use that a, b commute with x:

$$(a * b) * x = a * (b * x) = a * (x * b) = (a * x) * b = (x * a) * b = x * (a * b).$$

Since this holds for every $x \in S$, the element $a * b$ belongs to H. No commutativity of the whole set is assumed; we only used the defining property of these two elements.

Exercise 2.37: Closure of idempotents

Take $a, b \in H$, so $a * a = a$ and $b * b = b$. Associativity and commutativity allow regrouping as

$$(a * b) * (a * b) = (a * a) * (b * b) = a * b.$$

For justification, first write $a*(b*(a*b))$, commute b past a inside, then reassociate into $(a*a)*(b*b)$. Thus $a * b$ is idempotent and lies in H.

3 Isomorphic Binary Structures

Exercise 3.1: The three isomorphism checks

Check that the map is (1) one-to-one, (2) onto the specified target set, and (3) operation-preserving: $\phi(a * b) = \phi(a) *' \phi(b)$ for every $a, b \in S$. The first two make a bijection; the third ensures that it respects the algebraic structure. Also confirm initially that the formula actually takes values in the stated codomain.

Exercise 3.2: Negation on the integers

This is an isomorphism. If $-m = -n$ then $m = n$, proving injectivity. Each integer y is the image of $-y$, proving surjectivity. Finally $\phi(m+n) = -(m+n) = -m-n = \phi(m)+\phi(n)$, proving preservation of addition.

Exercise 3.3: Doubling integers

This map preserves addition since $2(m + n) = 2m + 2n$ and is injective because $2m = 2n$ implies $m = n$. However its image consists only of even integers; 1 has no preimage in $\mathbb{Z}$. Thus it is not onto and is not an isomorphism onto the specified target.

Exercise 3.4: Translating integers

The map $n \mapsto n + 1$ is bijective, with inverse $y \mapsto y - 1$, but does not preserve addition. Indeed $\phi(0 + 0) = 1$, while $\phi(0) + \phi(0) = 2$. A bijection alone is not an isomorphism.

Exercise 3.5: Halving rational numbers

If $x/2 = y/2$, then $x = y$, so the map is injective. A rational target q has rational preimage $2q$, so it is onto. Also $(x + y)/2 = x/2 + y/2$. All three requirements hold, making this an additive isomorphism.

Exercise 3.6: Squaring rational numbers

Squaring preserves multiplication: $(xy)^2 = x^2y^2$. It is not injective, since 1 and -1 have the same image, and is not onto $\mathbb{Q}$, since no rational square equals -1. Hence it is not an isomorphism.

Exercise 3.7: Cubing real numbers

The cubic function is strictly increasing, so it is injective, and every real y has real preimage $\sqrt[3]{y}$, so it is onto. Since $(xy)^3 = x^3y^3$, it preserves multiplication. Thus it is an isomorphism of the indicated multiplicative binary structures, even though the structures include zero and are not groups.

Exercise 3.8: Determinants of two by two matrices

Determinants preserve multiplication: $\det(AB) = \det(A)\det(B)$. The map is onto since $\det \operatorname{diag}(r, 1) = r$. It is not injective: the distinct matrices I_2 and $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ both have determinant 1. Thus it is not an isomorphism.

Exercise 3.9: Determinants of one by one matrices

A one by one matrix $[r]$ has determinant r. The map $[r] \mapsto r$ is bijective, with inverse $r \mapsto [r]$. Also $[r][s] = [rs]$, whose determinant is rs. Consequently this determinant map is an isomorphism.

Exercise 3.10: A decreasing exponential

The codomain here is $\mathbb{R}^+$, the positive reals. The function $\phi(r) = (1/2)^r$ is strictly decreasing and has inverse $y \mapsto \log_{1/2} y$ for $y > 0$, so it is bijective. The exponent law gives $\phi(r+s) = (1/2)^{r+s} = \phi(r)\phi(s)$. Hence it is an isomorphism from addition to multiplication.

Exercise 3.11: Differentiating smooth functions

Differentiation preserves addition: $(f + g)' = f' + g'$. It is onto the smooth functions: if h is smooth, $f(x) = \int_0^x h(t)\,dt$ is smooth and has derivative h. But it is not injective, since the constant functions 0 and 1 both map to zero. Thus it is not an isomorphism.

Exercise 3.12: Evaluating a derivative at zero

The map is additive because $(f + g)'(0) = f'(0) + g'(0)$. It is onto $\mathbb{R}$, since $f(x) = cx$ has $f'(0) = c$ for any real c. It is not injective: distinct constant functions have the same image 0. Therefore it is not an isomorphism.

Exercise 3.13: Integration from zero

Integration is additive, and if $\int_0^x f(t)\,dt = \int_0^x g(t)\,dt$ for all x, differentiation gives $f = g$, proving injectivity. However each image function is zero at $x = 0$, so the constant function 1 is not in the image. Thus it is not onto F and is not an isomorphism.

Exercise 3.14: Differentiating the integral

The fundamental theorem of calculus gives $\phi(f)(x) = \frac{d}{dx}\int_0^x f(t)\,dt = f(x)$, since every smooth function is continuous. Thus the map is the identity on F, so is bijective and preserves addition. It is an isomorphism.

Exercise 3.15: Multiplication by the independent variable

The map is not onto: every image $xf(x)$ vanishes at zero, whereas the constant function 1 does not. It also fails to preserve multiplication: taking $f = g = 1$ gives $\phi(fg)(x) = x$ but $(\phi(f)\phi(g))(x) = x^2$, different at $x = 2$. Thus it is not an isomorphism for either reason.

Exercise 3.16: Transporting addition by a translation

The inverse of $\phi(n) = n + 1$ is $\phi^{-1}(u) = u - 1$.

(a) To send ordinary addition to $*$, define $u * v = \phi((u - 1) + (v - 1)) = u + v - 1$. Then $\phi(m) * \phi(n) = (m + 1) + (n + 1) - 1 = m + n + 1 = \phi(m + n)$. Its identity is $\phi(0) = 1$.

(b) To send $*$ to ordinary addition, require $\phi(m*n) = \phi(m)+\phi(n)$. This gives $m*n+1 = m+n+2$, or $m * n = m + n + 1$. Its identity is $-1 = \phi^{-1}(0)$.

Exercise 3.17: Transporting multiplication by a translation

Use $\phi(n) = n + 1$ and its inverse $u - 1$.

(a) Define $u * v = \phi((u - 1)(v - 1)) = (u - 1)(v - 1) + 1 = uv - u - v + 2$. Then $\phi(m) * \phi(n) = mn + 1 = \phi(mn)$. The identity is $\phi(1) = 2$.

(b) Require $\phi(m * n) = \phi(m)\phi(n)$, giving $m * n = (m + 1)(n + 1) - 1 = mn + m + n$. Its identity is $0 = \phi^{-1}(1)$.

These are binary operations on integers; the construction does not assert that every integer has a multiplicative inverse.

Exercise 3.18: Transporting rational addition

Here $\phi(x) = 3x - 1$ and $\phi^{-1}(u) = (u + 1)/3$.

(a) Define $u*v = \phi((u+1)/3+(v+1)/3) = u+v+1$. This satisfies $\phi(x)*\phi(y) = 3x+3y-1 = \phi(x+y)$. Its identity is $-1 = \phi(0)$.

(b) Define $x * y = \phi^{-1}(\phi(x) + \phi(y)) = (3x + 3y - 2 + 1)/3 = x + y - 1/3$. Its identity is $1/3 = \phi^{-1}(0)$.

All displayed formulas preserve rationality, so their domains and codomains are correct.

Exercise 3.19: Transporting rational multiplication

Again $\phi^{-1}(u) = (u + 1)/3$.

(a) Define $u * v = \phi(\phi^{-1}(u)\phi^{-1}(v)) = (u + 1)(v + 1)/3 - 1$. Then $\phi(x) * \phi(y) = 3xy - 1 = \phi(xy)$. Its identity is $\phi(1) = 2$.

(b) Define $x * y = \phi^{-1}(\phi(x)\phi(y)) = ((3x - 1)(3y - 1) + 1)/3 = 3xy - x - y + 2/3$. Its identity is $\phi^{-1}(1) = 2/3$.

In each direction the equation used to define the operation is exactly the required preservation condition, and the original map is already bijective.

Exercise 3.20: Commuting with the operations

There are two routes from an input pair (a, b) to an element of S'. One first computes $a * b$ in S and then applies ϕ. The other first applies ϕ to both inputs and then computes $\phi(a) *' \phi(b)$ in S'. Saying that ϕ commutes with these operations means precisely that the two routes give the same result. It does not mean ϕ itself is a binary operation.

Exercise 3.21: Completing the isomorphism definition

Let $*$ and $*'$ be binary operations on S and S'. An isomorphism $\phi : (S, *) \to (S', *')$ is a bijection such that $\phi(a * b) = \phi(a) *' \phi(b)$ for every $a, b \in S$. The original version omitted bijectivity and the universal quantifier.

Exercise 3.22: The identity definition

The stated definition is correct: an element $e \in S$ is an identity for $*$ when $s * e = s = e * s$ for every $s \in S$. The middle equality to s says multiplication by e leaves every element unchanged, and the two sides specify both a left and a right identity. Merely requiring $s * e = e * s$, without equality to s, would only require e to commute with every element and would not suffice.

Exercise 3.23: Synopsis of identity uniqueness

Compute the product of two proposed identities: using the first as a left identity and the second as a right identity makes that same product equal to each of them, forcing them to coincide.

Exercise 3.24: Left and right identities

(a) A left identity $e_L \in S$ satisfies $e_L * s = s$ for every $s \in S$. It need not be unique: on $S = \{a, b\}$ define $x * y = y$. Then both elements are left identities. In the two-sided uniqueness proof, the step $e_L * e'_L = e_L$ is unavailable: it would require e'_L to be a right identity.

(b) A right identity $e_R \in S$ satisfies $s * e_R = s$ for every $s \in S$. On $S = \{a, b\}$ define $x * y = x$. Then both elements are right identities. The step $e_R * e'_R = e'_R$ fails because it requires e_R to be a left identity.

Exercise 3.25: A left identity and a right identity together

They cannot differ. If l is a left identity and r a right identity, then $l * r = r$ by the left identity property and $l * r = l$ by the right identity property. Thus $l = r$, and this element is consequently a two-sided identity. No associativity assumption is needed.

Exercise 3.26: The inverse of an isomorphism

The inverse ϕ^{-1} is a bijection because ϕ is. Given $u, v \in S'$, write $a = \phi^{-1}(u)$ and $b = \phi^{-1}(v)$. Preservation by ϕ gives $\phi(a * b) = \phi(a) *' \phi(b) = u *' v$. Applying ϕ^{-1} yields

$$\phi^{-1}(u *' v) = a * b = \phi^{-1}(u) * \phi^{-1}(v).$$

Thus the inverse also preserves the operations, completing all requirements.

Exercise 3.27: Composing isomorphisms

A composite of two bijections is a bijection (its inverse is $\phi^{-1} \circ \psi^{-1}$). Also

$$(\psi \circ \phi)(a * b) = \psi(\phi(a) *' \phi(b)) = \psi(\phi(a)) *'' \psi(\phi(b)).$$

The first preservation step uses ϕ and the second uses ψ. This is precisely preservation by $\psi \circ \phi$, so the composite is an isomorphism.

Exercise 3.28: Isomorphism is an equivalence relation

Reflexivity: the identity map on any binary structure is bijective and operation-preserving. Symmetry: if one structure is isomorphic to another, the inverse isomorphism from Exercise 3.26 gives the reverse direction. Transitivity: compose the two isomorphisms using Exercise 3.27. Thus all three equivalence-relation conditions hold on any specified set of binary structures.

Exercise 3.29: Commutativity is structural

Suppose $\phi : S \to S'$ is an isomorphism and $*$ is commutative. For arbitrary $u, v \in S'$, surjectivity gives $u = \phi(a)$ and $v = \phi(b)$. Then

$$u *' v = \phi(a * b) = \phi(b * a) = v *' u.$$

Hence $*'$ is commutative. Applying the same argument to ϕ^{-1} proves the converse. Therefore commutativity depends on structure, not on the names of elements.

Exercise 3.30: Associativity is structural

For arbitrary $u = \phi(a), v = \phi(b), w = \phi(c)$ in S', preservation gives

$$(u *' v) *' w = \phi((a * b) * c) = \phi(a * (b * c)) = u *' (v *' w).$$

Surjectivity ensures every target triple has these preimages; associativity in S gives the middle equality. Repeat with the inverse to obtain the reverse implication. Thus associativity is invariant under isomorphism.

Exercise 3.31: Existence of square roots is structural

Assume each $c \in S$ has some x with $x * x = c$. Given $d \in S'$, choose $c = \phi^{-1}(d)$ and a square root x of c. Then $y = \phi(x)$ satisfies $y *' y = \phi(x * x) = \phi(c) = d$. The same argument using ϕ^{-1} proves the converse. We require existence, not uniqueness, of these roots.

Exercise 3.32: Existence of an idempotent is structural

If $b * b = b$ in S, then $\phi(b) *' \phi(b) = \phi(b * b) = \phi(b)$, so $\phi(b)$ is idempotent in S'. Conversely any target idempotent pulls back under ϕ^{-1} to an idempotent in S. Thus the existence of such an element is a structural property.

Exercise 3.33: Complex numbers as real matrices

Define $\phi(a+bi) = M(a,b) = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$. Equality of matrices recovers both a and b, proving injectivity, and every matrix in H has this form, proving surjectivity.

(a) $M(a,b) + M(c,d) = M(a+c, b+d) = \phi((a+bi)+(c+di))$, so the map preserves addition.

(b) $M(a,b)M(c,d) = M(ac-bd, ad+bc) = \phi((a+bi)(c+di))$, so it preserves multiplication.

The same bijection therefore gives both requested isomorphisms.

Exercise 3.34: Classifying two-element binary structures

Encode a table on $\{a,b\}$ by the four entries $(a*a, a*b, b*a, b*b)$. The only relabelings are the identity and the swap of a, b. Here are ten representative tables, specified by their four entries in that order:

	$a*a$	$a*b$	$b*a$	$b*b$
1	a	a	a	a
2	a	a	a	b
3	a	a	b	a
4	a	a	b	b
5	a	b	a	a
6	a	b	a	b
7	a	b	b	a
8	b	a	a	a
9	b	a	b	a
10	b	b	a	a

To see completeness without guessing, encode a, b as $0, 1$. A swap sends (t_1, t_2, t_3, t_4) to $(1-t_4, 1-t_3, 1-t_2, 1-t_1)$. Exactly four tables are fixed: choose t_1, t_2 freely and then $t_4 = 1-t_1, t_3 = 1-t_2$. The other twelve tables form six pairs. Hence there are $4 + 12/2 = 10$ classes; the ten displayed representatives lie in distinct swap classes and exhaust them.

4 Groups

Exercise 4.1: Integer multiplication

Integer multiplication is associative ($\mathcal{G}_1$) and has identity 1 ($\mathcal{G}_2$). But not every integer has a multiplicative inverse in $\mathbb{Z}$: $0x = 1$ is impossible, and $2x = 1$ has no integer solution. Thus the first failing axiom is $\mathcal{G}_3$, so this is not a group.

Exercise 4.2: Even integers under addition

The sum $2m + 2n = 2(m + n)$ is even, so closure holds. Addition is associative, $0 = 2 \cdot 0$ is the identity, and the inverse of $2m$ is $-2m$, again even. All three group axioms hold, so this is an abelian group.

Exercise 4.3: The geometric-mean operation

Associativity already fails, so $\mathcal{G}_1$ is the first failing axiom. For example, on $\mathbb{R}^+$ let $a = 1, b = 1, c = 16$. Then $(a * b) * c = 1 * 16 = 4$, whereas $a * (b * c) = 1 * 4 = 2$. The positive square roots make this a binary operation, but not a group.

Exercise 4.4: Rational multiplication including zero

Multiplication is associative and has identity 1. Zero has no inverse, since $0q = 0$ for every rational q. Thus $\mathcal{G}_3$ is the first failure and the set $\mathbb{Q}$, as opposed to $\mathbb{Q} \setminus \{0\}$, is not a multiplicative group.

Exercise 4.5: Division as an operation

The operation is closed on nonzero reals, but is not associative. With $a = 1, b = 1, c = 2$, $(a * b) * c = (1/1)/2 = 1/2$, whereas $a * (b * c) = 1/(1/2) = 2$. Therefore the first failing axiom is $\mathcal{G}_1$.

Exercise 4.6: The absolute value of a product

Associativity holds: $(a * b) * c = \big||ab|c\big| = |a||b||c| = \big|a|bc|\big| = a * (b * c)$. However no identity exists in $\mathbb{C}$. Any product $a * e = |ae|$ is a nonnegative real number and so cannot equal, for example, $a = i$. Thus $\mathcal{G}_2$ is the first failure.

Exercise 4.7: An abelian group of order one thousand

Take $\mathbb{Z}_{1000}$ under addition modulo 1000. It has exactly 1000 elements, addition of residues is associative and commutative, zero is the identity, and k has inverse the representative of $-k$ modulo 1000. Hence this example meets every requirement.

Exercise 4.8: Units modulo eight

Multiply each row label by each column label and reduce modulo 8:

$\cdot_8$	1	3	5	7
1	1	3	5	7
3	3	1	7	5
5	5	7	1	3
7	7	5	3	1

The table shows closure, identity 1, and that each element is its own inverse. Associativity is inherited from integer multiplication modulo 8. Thus this is an abelian group (the four-element noncyclic group).

Exercise 4.9: Distinguishing the circle group

In U the equation $z^3 = 1$ has three solutions, the three cube roots of unity. In the additive group $\mathbb{R}$, the corresponding equation $3x = 0$ has only $x = 0$. In the multiplicative group $\mathbb{R}^*$, $x^3 = 1$ has only $x = 1$. An isomorphism sends the identity to the identity and preserves products and powers, so would biject these solution sets. Their different sizes rule out both proposed isomorphisms.

Exercise 4.10: Multiples of a positive integer

(a) Sums of multiples of n are multiples of n, addition is associative, $0 = n \cdot 0$ belongs, and the negative of nm is $n(-m)$. Thus $(n\mathbb{Z}, +)$ is a group.

(b) Define $\phi : n\mathbb{Z} \to \mathbb{Z}$ by $\phi(nm) = m$. This is well defined since $n \neq 0$ makes the representation unique; it is bijective with inverse $m \mapsto nm$. Finally $\phi(nm + nk) = m + k = \phi(nm) + \phi(nk)$, so it is an isomorphism.

Exercise 4.11: Diagonal matrices under addition

This is a group. Adding diagonal matrices adds their diagonal entries and leaves off-diagonal entries zero. Associativity follows from matrix addition; the zero matrix is the identity; and negating every diagonal entry gives an additive inverse still in the set. The group is abelian.

Exercise 4.12: All diagonal matrices under multiplication

The set is closed, multiplication is associative, and I_n is its identity, but the zero matrix has no multiplicative inverse. Therefore it is not a group. Requiring nonzero diagonal entries would remove this obstruction.

Exercise 4.13: Invertible diagonal matrices

The product of $\operatorname{diag}(a_1, \ldots, a_n)$ and $\operatorname{diag}(b_1, \ldots, b_n)$ is $\operatorname{diag}(a_1b_1, \ldots, a_nb_n)$ with nonzero diagonal. The identity is I_n, the inverse is $\operatorname{diag}(1/a_1, \ldots, 1/a_n)$, and multiplication is associative. Thus these matrices form an abelian group.

Exercise 4.14: Diagonal sign matrices

A product of two signs is a sign, so these diagonal matrices are closed. The identity has all diagonal entries 1, and each matrix is its own inverse since $(\pm 1)^2 = 1$. Associativity is inherited from matrix multiplication. Hence this is a group of 2^n elements.

Exercise 4.15: All upper-triangular matrices multiplicatively

Products remain upper triangular and I_n belongs, but singular upper-triangular matrices are also included. In particular the zero matrix cannot have a multiplicative inverse. Thus the given set is not a group under multiplication.

Exercise 4.16: Upper-triangular matrices additively

Sums and negatives remain upper triangular. Matrix addition is associative, the zero matrix is an additive identity in this set, and $-A$ is the additive inverse of A. Therefore these matrices form an abelian group under addition.

Exercise 4.17: Upper-triangular matrices of determinant one

If A, B are upper triangular with determinant 1, so is AB, since $\det(AB) = 1$. The identity belongs. Each A is invertible, and A^{-1} is upper triangular: solving $Ax = e_j$ by back substitution gives zeros below the jth position in column j of the inverse. Also $\det(A^{-1}) = 1$. Associativity is inherited, so this is a group.

Exercise 4.18: Matrices of determinant plus or minus one

The determinant of a product is the product of determinants, so closure holds in $\{1, -1\}$. Every such matrix is invertible, I_n belongs, and $\det(A^{-1}) = 1/\det A \in \{1, -1\}$. Associativity is matrix associativity, so the indicated set is a group.

Exercise 4.19: A shifted multiplicative group

Use the identity $a * b + 1 = (a + 1)(b + 1)$.

(a) If $a, b \neq -1$, their shifted values are nonzero, so $(a * b) + 1 \neq 0$ and $a * b \neq -1$. The formula specifies one real output, proving it is a binary operation on S.

(b) The bijection $\phi : S \to \mathbb{R}^*$, $\phi(a) = a + 1$, satisfies $\phi(a * b) = \phi(a)\phi(b)$, so it transports the group laws from $\mathbb{R}^*$. Explicitly, both bracketings have shifted value $(a + 1)(b + 1)(c + 1)$, proving associativity; the identity is 0, and the inverse of a is $-a/(a+1)$, whose shifted value is $1/(a+1)$.

(c) Apply ϕ to $2 * x * 3 = 7$: $3(x + 1)4 = 8$. Thus $x + 1 = 2/3$ and $x = -1/3 \in S$. Checking directly, $2 * (-1/3) = 1$ and $1 * 3 = 7$.

Exercise 4.20: The groups of order four

No repeated entry can occur in a row or column of a group table, by cancellation. Filling accordingly gives the following three tables in the order e, a, b, c:

T_1	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

T_2	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	a	e
c	c	b	e	a

T_3	e	a	b	c
e	e	a	b	c
a	a	b	c	e
b	b	c	e	a
c	c	e	a	b

The first two have $a^2 = e$; the third has $a^2 = b$. In T_2, b has successive powers e, b, a, c, whereas in T_3, a has successive powers e, a, b, c. Therefore $T_2 \cong T_3$ via $e \mapsto e, a \mapsto b, b \mapsto a, c \mapsto c$. In T_1 every nonidentity element squares to e, so it cannot be isomorphic to either cyclic table.

(a) Yes: all three tables are symmetric, so every group of order four is abelian.

(b) T_3 is U_4 via $e \mapsto 1, a \mapsto i, b \mapsto -1, c \mapsto -i$; T_2 is also U_4 via its preceding relabeling.

(c) T_1 is the diagonal-sign group for $n = 2$: send $a \mapsto \mathrm{diag}(-1, 1)$, $b \mapsto \mathrm{diag}(1, -1)$, $c \mapsto -I_2$. Multiplying these matrices verifies the table and therefore its associativity.

Exercise 4.21: Counting labeled small group operations

On a fixed two-element set there are two group operations: either element can be the identity, and that choice forces the other element to square to it. On a fixed three-element set there are three operations, one for each choice of identity; after that choice the two other elements must be inverses and their squares are forced. Swapping those two nonidentity labels leaves the same cyclic group table, so does not create an additional operation. Thus the answers are 2 and 3.

Exercise 4.22: Ordering the group axioms

The inverse axiom $\mathcal{G}_3$ refers to the identity e introduced in $\mathcal{G}_2$. Thus $\mathcal{G}_2$ must precede $\mathcal{G}_3$. The unacceptable orders are $\mathcal{G}_1\mathcal{G}_3\mathcal{G}_2$, $\mathcal{G}_3\mathcal{G}_1\mathcal{G}_2$, and $\mathcal{G}_3\mathcal{G}_2\mathcal{G}_1$. The three orders with $\mathcal{G}_2$ first relative to $\mathcal{G}_3$ are acceptable; associativity has no such dependence on the other two.

Exercise 4.23: Critiquing four proposed definitions

A correct definition specifies a binary operation on G, associativity, an element $e \in G$ satisfying $ex = xe = x$ for every x, and for each $a \in G$ an $a^{-1} \in G$ satisfying $aa^{-1} = a^{-1}a = e$.

(a) The equality ending "x = identity" is incorrect: x is arbitrary, not necessarily the identity. The identity equation needs "for every $x \in G$," the inverse must belong to G, and the same operation symbol must be used consistently.

(b) It refers to "the operation" without specifying a binary operation on G. It should define what the identity and inverse equations mean and explicitly put each inverse in G.

(c) Associativity is missing. "An inverse exists" must mean an inverse for every element, relative to the specified identity; merely defining a binary operation is already part of the setup.

(d) "Associative under addition" confuses the operation with another operation. The notation $\{e\}$ denotes a set, not the element e; the identity equation must end in a, not e. The inverse must be in G, and the universal quantifiers must govern the appropriate equations.

Exercise 4.24: Identity and inverses without associativity

Consider

$*$	e	a	b
e	e	a	b
a	a	e	e
b	b	e	e

The first row and column show e is a two-sided identity. Both a and b are their own two-sided inverses, so $\mathcal{G}_2, \mathcal{G}_3$ hold. But $(a * a) * b = e * b = b$, whereas $a * (a * b) = a * e = a$. Thus associativity fails, as requested.

Exercise 4.25: Ten statements about groups

(a) False: two identities e, f satisfy $e = ef = f$.

(b) True: the cancellation rule forces the unique three-element cyclic table, up to relabeling.

(c) True for the linear equations in this section: $ax = b$ and $ya = b$ have the solutions $a^{-1}b$ and ba^{-1}.

(d) False: understanding and stating the exact content matters; identical wording is unnecessary.

(e) False: one implication could allow a definition that excludes some actual groups. Equivalence requires both directions.

(f) True: equivalence of the two definitions is precisely what is needed.

(g) True: the one-, two-, and three-element group tables are all commutative.

(h) True: $axb = c$ has the unique solution $x = a^{-1}cb^{-1}$, obtained by multiplying on the indicated sides.

(i) False: a group must contain its identity, so it is nonempty.

(j) True: a group is a set with a binary operation satisfying additional axioms.

Exercise 4.26: Synopsis of left cancellation

Multiply an equality with the same left factor on the left by that factor's inverse, then reassociate and use the identity to remove that factor.

Exercise 4.27: Synopsis of solving a group equation

Multiplying the desired right-hand side on the left by the inverse of the coefficient produces a solution. Left cancellation shows that any two solutions must agree.

Exercise 4.28: Isomorphisms preserve inverses

Let a^{-1} be the inverse of a and let e' be the target identity. Since $\phi(e) = e'$,

$$\phi(a)\phi(a^{-1}) = \phi(aa^{-1}) = e', \qquad \phi(a^{-1})\phi(a) = \phi(a^{-1}a) = e'.$$

Thus $\phi(a^{-1})$ is a two-sided inverse of $\phi(a)$. The inverse of a group element is unique, so $\phi(a^{-1}) = \phi(a)^{-1}$.

Exercise 4.29: An involution in a group of even order

Pair each element x with its inverse. Elements with $x \neq x^{-1}$ occur in disjoint pairs and hence contribute an even number of elements. If e were the only self-inverse element, the total size would therefore be even plus one, which is odd. Since the group has even size, there must be some $a \neq e$ with $a = a^{-1}$, equivalently $a^2 = e$.

Exercise 4.30: Mixed one-sided axioms

On $\mathbb{R}^*$ put $a * b = |a|b$.

(a) Outputs are nonzero, and $(a * b) * c = ||a|b|c = |a||b|c = a * (b * c)$, proving associativity.

(b) The element 1 is a left identity, since $1 * b = b$. Each a has a right inverse relative to 1, namely $b = 1/|a|$, since $a * b = 1$.

(c) This is not a group. A right identity r would need $1 * r = 1$, hence $r = 1$, but then $(-1) * r = 1 \neq -1$.

(d) A left identity combined with right inverses does not suffice for the group axioms. The valid one-sided alternatives require the identity and inverse axioms to have matching sides (both left, or both right). This example even has two left identities, 1 and -1.

Exercise 4.31: Idempotents in a group

The identity is idempotent since $e^2 = e$. If $x^2 = x$, multiply on the left by x^{-1} to obtain $x = e$. Thus every idempotent equals the identity, and there is exactly one.

Exercise 4.32: A group of exponent two is abelian

Assume $x^2 = e$ for every $x \in G$. In particular $(ab)^2 = e$, so $abab = e$. Multiplying this equality on the left by a and on the right by b gives $ba = ab$, because $a^2 = b^2 = e$. Since a, b were arbitrary, the group is abelian. No commutation was used before establishing the conclusion.

Exercise 4.33: Powers of commuting products

Fix a, b in an abelian group. For $n = 1$, $(ab)^1 = a^1b^1$. If $(ab)^n = a^nb^n$, then

$$(ab)^{n+1} = (ab)^n ab = a^nb^nab = a^nab^nb = a^{n+1}b^{n+1}.$$

Commutativity permits moving a past b^n, and associativity permits regrouping. Thus mathematical induction proves the identity for every positive integer n.

Exercise 4.34: A finite-order power

Let $|G| = m$. The $m+1$ powers $e, a, a^2, \ldots, a^m$ cannot all be distinct, so $a^i = a^j$ for some $0 \le i < j \le m$. Multiply by a^{-i} to get $a^{j-i} = e$. Since $j - i > 0$, this supplies the required positive exponent.

Exercise 4.35: A squared-product identity forces commutation

The hypothesis is $abab = aabb$. Multiply on the left by a^{-1} to get $bab = abb$, then on the right by b^{-1} to get $ba = ab$. The sides of multiplication matter; cancellation is valid without assuming the group is abelian.

Exercise 4.36: When inverses keep their order

In any group $(ab)^{-1} = b^{-1}a^{-1}$, since this element gives e on multiplication on either side. If $ab = ba$, then their inverses also commute, so this equals $a^{-1}b^{-1}$. Conversely, if $(ab)^{-1} = a^{-1}b^{-1}$, comparison with the general formula gives $b^{-1}a^{-1} = a^{-1}b^{-1}$. Taking inverses reverses both products and yields $ab = ba$. This proves both implications.

Exercise 4.37: Cyclically rotating a product equal to the identity

From $abc = e$, multiply on the left by a^{-1} to get $bc = a^{-1}$. Multiplying on the right by a then gives $bca = a^{-1}a = e$. This is a cyclic rotation, not an arbitrary interchange of factors.

Exercise 4.38: Matching left axioms imply the group axioms

Assume associativity, a left identity e, and a left inverse a' for every a.

First obtain left cancellation.

If $ax = ay$, multiply on the left by a'. Then $(a'a)x = (a'a)y$ gives $ex = ey$, hence $x = y$.

Prove the identity is also right-sided.

Since $a'(ae) = (a'a)e = ee = e = a'a$, left cancellation of a' gives $ae = a$.

Prove each left inverse is also right-sided.

Now $a'(aa') = (a'a)a' = ea' = a' = a'e$. Cancel a' on the left to get $aa' = e$. Thus e is a two-sided identity and each a' is a two-sided inverse. All ordinary group axioms hold.

Exercise 4.39: Solvability of all linear equations

Choose $a \in G$, possible since G is nonempty. By the solvability hypothesis choose e with $ea = a$. For any $g \in G$, choose z with $az = g$. Associativity gives $eg = e(az) = (ea)z = az = g$, so e is a left

identity for every element. For any $c \in G$, solvability of $yc = e$ gives a left inverse c' with $c'c = e$. Thus the matching left axioms of Exercise 4.38 hold, and its proof shows that G is a group.

Exercise 4.40: The reversed operation

Write the original multiplication as juxtaposition and set $a * b = ba$. Then $(a * b) * c = cba = a * (b * c)$, the original identity remains an identity, and the original a^{-1} is also the $*$-inverse. Thus $(G, *)$ is a group. Define $\phi(a) = a^{-1}$. It is bijective because applying it twice gives a, and

$$\phi(a * b) = \phi(ba) = (ba)^{-1} = a^{-1}b^{-1} = \phi(a)\phi(b).$$

Hence it is an isomorphism from the reversed group to the original group.

Exercise 4.41: Conjugation is an automorphism

For fixed $g \in G$ define $i_g(x) = gxg^{-1}$. Its inverse map is $i_{g^{-1}}$, since $g^{-1}(gxg^{-1})g = x$ and the reverse composition also equals the identity. Thus i_g is bijective. Moreover

$$i_g(x)i_g(y) = (gxg^{-1})(gyg^{-1}) = gx(g^{-1}g)yg^{-1} = gxyg^{-1} = i_g(xy).$$

Therefore conjugation preserves multiplication and is an isomorphism of G with itself.

5 Subgroups

Exercise 5.1: Real numbers in the additive complex group

Yes. The real numbers contain 0, and for any reals a, b both $a + b$ and $-a$ are real. Associativity is inherited from complex addition. These facts verify the subgroup conditions.

Exercise 5.2: Positive rationals additively

No. The additive identity 0 is not positive, so it is not in $\mathbb{Q}^+$. Also a positive rational has a negative additive inverse, outside the set. Closure under addition alone is insufficient.

Exercise 5.3: Multiples of seven

Yes. The set $7\mathbb{Z}$ contains zero. If $7m, 7n \in 7\mathbb{Z}$, then $7m + 7n = 7(m + n)$ and $-(7m) = 7(-m)$ remain in the set. Thus it is an additive subgroup of $\mathbb{C}$.

Exercise 5.4: Pure imaginary numbers

Yes. The identity is $0 = i \cdot 0$. For real a, b, $ia + ib = i(a + b)$ and $-(ia) = i(-a)$ are pure imaginary. Therefore closure and inverse closure hold.

Exercise 5.5: Rational multiples of pi

Yes. The set is $\pi\mathbb{Q} = \{\pi q : q \in \mathbb{Q}\}$. It contains zero, sums have the form $\pi(q + r)$, and negatives have the form $\pi(-q)$. Since $q + r, -q$ are rational, all subgroup conditions hold under addition.

Exercise 5.6: Integral powers of pi additively

No. Every π^n is positive, so zero is absent and no negative additive inverse is included. Thus $\{\pi^n : n \in \mathbb{Z}\}$ is not an additive subgroup, regardless of whether particular sums happen to lie in it.

Exercise 5.7: The same six sets multiplicatively

The only multiplicative subgroups of $\mathbb{C}^*$ among the six are $\mathbb{Q}^+$ and $\{\pi^n : n \in \mathbb{Z}\}$. The positive rationals contain 1, are closed under products, and have positive rational reciprocals. Integral powers of π contain $\pi^0 = 1$, multiply by adding exponents, and invert by negating exponents. The other four sets $(\mathbb{R}, 7\mathbb{Z}, i\mathbb{R}, \pi\mathbb{Q})$ contain 0, so are not even subsets of the stated group $\mathbb{C}^*$.

Exercise 5.8: Matrices of determinant two

No. The identity matrix has determinant 1, so it is absent. In addition, the product of two matrices of determinant 2 has determinant 4, violating closure. Such matrices exist, for example $\mathrm{diag}(2, 1, \ldots, 1)$.

Exercise 5.9: Invertible diagonal matrices

Yes. The identity is diagonal, products multiply corresponding nonzero diagonal entries, and reciprocating those entries produces a diagonal inverse. Thus this nonempty subset is closed under products and inverses, hence is a subgroup of $GL(n, \mathbb{R})$.

Exercise 5.10: Invertible upper-triangular matrices

Yes. Products remain upper triangular and their diagonal entries are the corresponding products of nonzero diagonal entries. The identity belongs. Back substitution in $Ax = e_j$ shows A^{-1} is upper triangular, and its diagonal entries are $1/a_{jj}$, hence nonzero. Thus this is a subgroup.

Exercise 5.11: Matrices of determinant minus one

No. The identity has determinant 1, and the product of two determinant--1 matrices has determinant 1, outside the set. Hence both identity membership and product closure fail.

Exercise 5.12: Matrices of determinant one or minus one

Yes. The identity belongs, products have determinant in $\{1, -1\}$, and the inverse has determinant $1/\det A$, again in $\{1, -1\}$. All matrices in the set are invertible, so the subgroup criterion applies.

Exercise 5.13: Orthogonal matrices

The identity satisfies $I_n^T I_n = I_n$. If $A^T A = B^T B = I_n$, then $(AB)^T(AB) = B^T A^T AB = B^T B = I_n$, so products remain orthogonal. The equation $A^T A = I_n$ gives $A^{-1} = A^T$, and then $(A^{-1})^T A^{-1} = AA^T = I_n$. Thus inverses also belong, and this is a subgroup.

Exercise 5.14: Nowhere-zero functions

(a) Not an additive subgroup: the zero function is absent; also $1 + (-1) = 0$ shows failure of closure. **(b)** Yes, this is the entire multiplicative group $\widetilde{F}$. Products are nowhere zero, the constant 1 is the identity, and the inverse of f is the function $x \mapsto 1/f(x)$, defined everywhere because f never vanishes.

Exercise 5.15: Functions vanishing at one

(a) Yes. The zero function belongs, and if $f(1) = g(1) = 0$, then $(f + g)(1) = 0$ and $(-f)(1) = 0$. **(b)** No. Every function in the specified set vanishes at 1, so none belongs to $\widetilde{F}$. It is not a subset of the proposed ambient multiplicative group.

Exercise 5.16: Nowhere-zero functions equal to one at one

(a) No: the zero function is absent (and adding two members gives value 2 at 1). **(b)** Yes. The constant function 1 belongs, products have value $1 \cdot 1 = 1$ at 1, and reciprocals have value $1/1 = 1$

there. Nowhere-zero values are preserved by products and reciprocals.

Exercise 5.17: Nowhere-zero functions equal to one at zero

(a) No, since the additive identity is absent and adding two such functions gives value 2 at zero. **(b)** Yes: the constant 1 belongs, and both $(fg)(0) = f(0)g(0) = 1$ and $(1/f)(0) = 1$ hold. All resulting functions remain nowhere zero.

Exercise 5.18: Nowhere-zero functions equal to minus one at zero

(a) No: the zero function does not satisfy the defining condition. **(b)** No: the multiplicative identity has value 1, not -1, at zero. Furthermore multiplying two members gives value $(-1)(-1) = 1$ at zero, so product closure fails.

Exercise 5.19: Nonzero constant functions

(a) No: the zero constant is not in this subset of $\widetilde{F}$, and the constants $1, -1$ add to it. **(b)** Yes: nonzero constants contain 1, multiply to nonzero constants, and have nonzero constant reciprocals. Thus they form a multiplicative subgroup.

Exercise 5.20: All subgroup relations among nine groups

Include the nine reflexive relations $G_i \leq G_i$. All strict relations are the transitive consequences of

$$G_2 < G_8 < G_7 < G_1 < G_4, \qquad G_9 < G_3 < G_5, \qquad G_6 < G_5.$$

Explicitly the additional pairs (i, j) with $G_i < G_j$ are

$$(2,8),(2,7),(2,1),(2,4),(8,7),(8,1),(8,4),$$
$$(7,1),(7,4),(1,4),(9,3),(9,5),(3,5),(6,5).$$

The additive chain is $12\mathbb{Z} \subset 6\mathbb{Z} \subset 3\mathbb{Z} \subset \mathbb{Z} \subset \mathbb{R}$. Multiplicatively, powers of 6 are positive rationals, and powers of π are positive reals. The latter are not all rational, since π is irrational; the two cyclic power groups do not contain each other, since π is not a power of 6 and 6 is not an integral power of π (use $\pi < 6 < \pi^2$). A subgroup must use the ambient operation, so no cross-relations between the additive and multiplicative lists are allowed.

Exercise 5.21: Five elements of three cyclic groups

(a) In $25\mathbb{Z}$: $-50, -25, 0, 25, 50$, obtained using integer multiples $-2, -1, 0, 1, 2$ of 25.

(b) In $\{(1/2)^n : n \in \mathbb{Z}\}$: $4, 2, 1, 1/2, 1/4$, from those same exponents.

(c) In $\{\pi^n : n \in \mathbb{Z}\}$: $\pi^{-2}, \pi^{-1}, 1, \pi, \pi^2$.

Zero exponents in a multiplicative cyclic group give its identity 1; they never give the number 0.

Exercise 5.22: A signed interchange matrix

Let $A = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$. Direct multiplication gives $A^2 = I_2$, while $A \neq I_2$. Thus every even integral power is I_2 and every odd integral power is A (also for negative exponents, since $A^{-1} = A$). The subgroup is precisely $\{I_2, A\}$.

Exercise 5.23: A shear matrix

For $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, multiplication shows

$$\begin{pmatrix} 1 & m \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & m+n \\ 0 & 1 \end{pmatrix}.$$

Thus induction gives $A^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$ for positive n, and the inverse $A^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$ extends it to every integer n. These, and only these, are the elements of $\langle A \rangle$.

Exercise 5.24: A diagonal generator

If $A = \operatorname{diag}(3, 2)$, then $A^n = \operatorname{diag}(3^n, 2^n)$ for every $n \in \mathbb{Z}$; negative powers use reciprocals of the diagonal entries. Therefore

$$\langle A \rangle = \{\operatorname{diag}(3^n, 2^n) : n \in \mathbb{Z}\}.$$

The same exponent must occur in both positions; the subgroup does not allow them to be chosen independently.

Exercise 5.25: A scaled signed interchange

Write $B = \begin{pmatrix} 0 & -2 \\ -2 & 0 \end{pmatrix} = 2A$ with $A^2 = I_2$ from Exercise 5.22. Since scalar matrices commute with A, $B^n = 2^n A^n$. Thus its elements are

$$\{2^{2k} I_2 : k \in \mathbb{Z}\} \cup \left\{ 2^{2k+1} \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} : k \in \mathbb{Z} \right\}.$$

This includes negative powers; $B^2 = 4I_2$ shows that the group is infinite.

Exercise 5.26: Which six groups are cyclic?

G_1 : Cyclic, generated exactly by $1, -1$: the multiples of an integer a fill $\mathbb{Z}$ only if $|a| = 1$.

G_2 : Not cyclic. If $q \neq 0$ were an additive generator, $q/2$ could not be an integral multiple of q; $q = 0$ plainly cannot generate $\mathbb{Q}$.

G_3 : Not cyclic. If a positive rational generator differs from 1, replace it by its reciprocal if necessary so it is $q > 1$. Then the rational $(1 + q)/2$ lies strictly between 1 and q, while no integral power of q does. The generator 1 also fails.

G_4 : Cyclic, with precisely the generators $6, -6$, since division by 6 identifies this group with $\mathbb{Z}$.

G_5 : Cyclic, with precisely $6, 1/6$ as generators. If 6^k generates, there must be an integer m with $6^{km} = 6$, so $km = 1$ and $k = \pm 1$.

G_6 : Not cyclic. If $a + b\sqrt{2}$ generated it, expressing $1 = k(a + b\sqrt{2})$ with integers forces $kb = 0$ and $ka = 1$, by irrationality of $\sqrt{2}$. Thus $b = 0$, $a = \pm 1$, and its multiples cannot include $\sqrt{2}$.

Exercise 5.27: The subgroup generated by three modulo four

Successive multiples of 3 modulo 4 are $0, 3, 2, 1$, then 0 again. These four residues are distinct, so the generated subgroup has order 4.

Exercise 5.28: A nonidentity element of the Klein group

Table 5.11 gives $c^2 = e$ and $c \neq e$. Therefore the subgroup generated by c is $\{e, c\}$ and has order 2.

Exercise 5.29: An angle of two pi over three

The generator is $e^{2\pi i/3}$. Its kth power is 1 exactly when $2\pi k/3$ is an integral multiple of 2π, or $3 \mid k$. The least positive such k is 3, so the subgroup has order 3, even though the ambient group has six elements.

Exercise 5.30: An angle of four pi over five

The generator is $e^{4\pi i/5}$. Its kth power is 1 if and only if $5 \mid 2k$. Since 2 and 5 are relatively prime, this is equivalent to $5 \mid k$. Thus the order is 5.

Exercise 5.31: An angle of three pi over two

The generator is $e^{3\pi i/2} = -i$. Its powers are $1, -i, -1, i$, then 1 again. Consequently its cyclic subgroup has order 4.

Exercise 5.32: An angle of five pi over four

The generator is $e^{5\pi i/4} = e^{2\pi i(5/8)}$. Its kth power is 1 exactly when $8 \mid 5k$. Because $\gcd(5, 8) = 1$, the least positive such k is 8. Thus this element generates all of U_8.

Exercise 5.33: A permutation matrix of order two

On the standard column vectors the matrix interchanges e_1, e_3 and interchanges e_2, e_4. Applying it twice fixes all four basis vectors, so $A^2 = I_4$. Since one application does not fix e_1, $A \neq I_4$. Therefore

its cyclic subgroup has order 2.

Exercise 5.34: A permutation matrix of order four

Track the first standard basis vector: $e_1 \mapsto e_3 \mapsto e_2 \mapsto e_4 \mapsto e_1$. This is a cycle containing all four basis vectors. Four applications fix each vector, and fewer than four do not fix e_1. Thus $A^4 = I_4$ with no smaller positive identity power, giving order 4.

Exercise 5.35: A permutation matrix of order three

This matrix sends $e_1 \mapsto e_4 \mapsto e_2 \mapsto e_1$ and fixes e_3. Thus three applications fix every standard basis vector. Neither one nor two applications fix e_1, so the least positive identity power is 3, and the generated subgroup has order 3.

Exercise 5.36: All subgroups generated in Z six

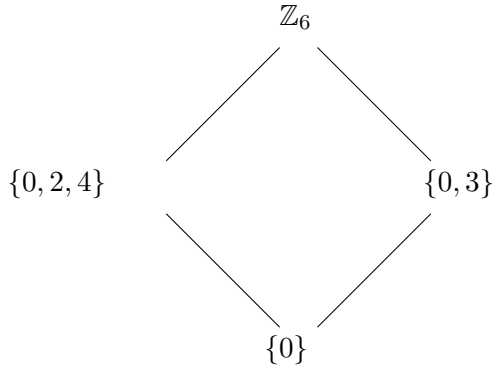
(a) Add row and column labels modulo 6:

$+$	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

(b) $\langle 0 \rangle = \{0\}$, $\langle 1 \rangle = \langle 5 \rangle = \mathbb{Z}_6$, $\langle 2 \rangle = \langle 4 \rangle = \{0, 2, 4\}$, $\langle 3 \rangle = \{0, 3\}$. These follow by repeatedly adding the generator until zero reappears.

(c) Exactly 1 and 5 generate all six residues.

(d) The inclusion diagram is the diamond below. Neither middle subgroup contains the other, since $2 \notin \{0, 3\}$ and $3 \notin \{0, 2, 4\}$.



Exercise 5.37: A complete subgroup definition

A subgroup H of a group G is a subset that is itself a group under the operation inherited from G. Equivalently, it contains e, is closed under products, and contains the inverse of each of its elements. The proposed definition omits product closure. For instance $\{0, 1, -1\}$ in the additive group $\mathbb{Z}$ contains the identity and all negatives but is not closed under addition.

Exercise 5.38: The definition of cyclic

The stated definition is correct: there exists $a \in G$ such that every element of G is a^n for some integer n. Both negative powers and $n = 0$ are allowed. In additive notation the same condition is $G = \{na : n \in \mathbb{Z}\}$.

Exercise 5.39: Ten subgroup statements

(a) True: associativity is one of the group axioms.

(b) False: multiplication by inverses proves both cancellation laws in every group.

(c) True: the inherited operation makes G a subgroup of itself.

(d) False: the text calls only G itself the improper subgroup; $\{e\}$ is the trivial subgroup, usually proper when $G \neq \{e\}$.

(e) False: in $\mathbb{Z}_4$, the element 2 generates only $\{0, 2\}$.

(f) False: $1, -1$ are distinct generators of the infinite cyclic group $\mathbb{Z}$.

(g) False: $\mathbb{Z}$ is a group under addition but zero has no multiplicative inverse.

(h) False as a proposed complete definition: a subgroup is a subset satisfying additional operation and group conditions.

(i) True: 1 generates $\mathbb{Z}_4$ by addition modulo 4.

(j) False: $\{1\}$ is a subset of the additive group $\mathbb{Z}$ but is not a subgroup, since it lacks zero.

Exercise 5.40: More than two square roots of the identity

In the Klein four-group $V = \{e, a, b, c\}$ every element satisfies $x^2 = e$. Thus there are four solutions, exceeding two. The familiar degree bound for polynomial roots over a field cannot be applied to an arbitrary group equation.

Exercise 5.41: An isomorphism carries subgroups to subgroups

Since H contains e, its image contains $\phi(e) = e'$. If $u = \phi(h)$ and $v = \phi(k)$ lie in $\phi[H]$, then $u *' v = \phi(h * k)$ belongs to $\phi[H]$ because $h * k \in H$. Also $u^{-1} = \phi(h^{-1})$ belongs because $h^{-1} \in H$. Thus the image contains the identity and is closed under products and inverses, so is a subgroup of G'.

Exercise 5.42: An isomorphism carries a cyclic group to a cyclic group

Let $G = \langle a \rangle$. Every $g' \in G'$ is $\phi(g)$ for some $g \in G$, by surjectivity. Write $g = a^n$. A homomorphism preserves positive powers by repeated multiplication, zero powers because it preserves the identity, and negative powers because it preserves inverses. Hence $g' = \phi(a)^n$. Thus $G' = \langle \phi(a) \rangle$ is cyclic.

Exercise 5.43: Products of two subgroups in an abelian group

Let $HK = \{hk : h \in H, k \in K\}$. It contains $ee = e$. If $h_1k_1, h_2k_2 \in HK$, then by commutativity

$$(h_1k_1)(h_2k_2) = (h_1h_2)(k_1k_2) \in HK.$$

Also $(hk)^{-1} = k^{-1}h^{-1} = h^{-1}k^{-1} \in HK$. The factors remain in their respective subgroups. Therefore HK is a subgroup; commutativity was essential in regrouping the factors.

Exercise 5.44: The hidden nonemptiness assumption

The argument starts "let $a \in H$," which is not possible when $H = \varnothing$. The empty subset is vacuously closed under products and inverses, so Conditions 1 and 3 alone do not force identity membership. If one separately assumes H nonempty, the argument becomes valid. The flaw is therefore not the algebraic calculation, but the unjustified choice of an element.

Exercise 5.45: The one-step subgroup criterion

If H is a subgroup, then $b^{-1} \in H$ and hence $ab^{-1} \in H$ for $a, b \in H$. Conversely assume $H \neq \varnothing$ and that all such products belong. Choose $h \in H$; then $hh^{-1} = e \in H$. For $b \in H$ use $a = e$ to get $b^{-1} \in H$. Finally for $a, b \in H$ the element b^{-1} also belongs, so $a(b^{-1})^{-1} = ab \in H$. Thus identity, inverse closure, and product closure all follow.

Exercise 5.46: Only one generator

If $G = \langle a \rangle$, then a^{-1} is also a generator because $(a^{-1})^{-n} = a^n$. With only one generator we must have $a = a^{-1}$, so $a^2 = e$. Every power of a is then either e or a, giving $|G| \leq 2$. This includes the trivial group, whose sole generator is its identity.

Exercise 5.47: The solutions of x squared equals identity

Let $H = \{x : x^2 = e\}$ in an abelian group. The identity belongs. If $x, y \in H$, commutativity gives $(xy)^2 = x^2y^2 = e$. Also $(x^{-1})^2 = (x^2)^{-1} = e$. Thus H is closed under products and inverses and is a subgroup. Without commutativity, the regrouping of $(xy)^2$ would not be justified.

Exercise 5.48: The solutions of a fixed power equation

For fixed $n \geq 1$, put $H = \{x : x^n = e\}$. The identity is in H. If $x, y \in H$, the commuting-product power law gives $(xy)^n = x^ny^n = e$, and $(x^{-1})^n = (x^n)^{-1} = e$. Hence the subgroup criterion holds. For $n = 1$ this is simply the trivial subgroup.

Exercise 5.49: A positive identity power in a finite group

The powers $e, a, a^2, \ldots, a^{|G|}$ form $|G|+1$ elements counted with repetitions in a set of size $|G|$. Thus $a^i = a^j$ for some $i < j$. Multiplying by a^{-i} gives $a^{j-i} = e$ with positive exponent, as required.

Exercise 5.50: A finite nonempty product-closed subset

Choose $h \in H$. By closure all positive powers of h are in H, and finiteness forces $h^i = h^j$ for some positive $i < j$. In the ambient group cancel to get $h^{j-i} = e \in H$. For any $a \in H$, the same finite-power argument produces $a^m = e$ with $m \geq 1$. If $m = 1$, then $a = e$ and its inverse is in H; otherwise $a^{-1} = a^{m-1} \in H$. Thus H contains the identity and all inverses, and the given product closure completes the subgroup conditions.

Exercise 5.51: The centralizer of one element

The identity commutes with a. If $xa = ax$ and $ya = ay$, then $(xy)a = x(ya) = x(ay) = (xa)y = a(xy)$. To obtain inverse closure, multiply $xa = ax$ on the left and right by x^{-1} to get $ax^{-1} = x^{-1}a$. Thus H_a contains e and is closed under products and inverses, making it a subgroup.

Exercise 5.52: The centralizer of a set and the center

(a) The identity commutes with every $s \in S$. If x, y do so, then for every such s, $(xy)s = x(ys) = x(sy) = (xs)y = s(xy)$. Similarly $xs = sx$ implies $x^{-1}s = sx^{-1}$. Hence H_S is a subgroup. If S is empty the condition is vacuous and $H_S = G$, still a subgroup.

(b) For $S = G$, take $x, y \in H_G$. Since x commutes with *every* element of G, it commutes in particular with y. Thus the subgroup H_G is abelian.

Exercise 5.53: An equivalence relation from a subgroup

Reflexivity follows from $aa^{-1} = e \in H$. If $ab^{-1} \in H$, its inverse ba^{-1} belongs to H, proving symmetry. If $ab^{-1}, bc^{-1} \in H$, their product is $ab^{-1}bc^{-1} = ac^{-1} \in H$, proving transitivity. These are all three required conditions; no commutativity or normality is assumed.

Exercise 5.54: Intersecting two subgroups

Both H and K contain e, so their intersection is nonempty. If $x, y \in H \cap K$, their product lies in H and lies in K, so lies in the intersection. The same applies to x^{-1} because both subgroups are inverse-closed. Associativity is inherited from G, and the subgroup criterion therefore proves $H \cap K \leq G$.

Exercise 5.55: A cyclic group is abelian

Write $G = \langle a \rangle$. Given $x, y \in G$, there are integers r, s with $x = a^r, y = a^s$. Then $xy = a^{r+s} = a^{s+r} = yx$, using commutativity of integer addition, not a preexisting commutativity assumption about G. Thus all elements of G commute.

Exercise 5.56: The set of all nth powers

A sufficient hypothesis is that G is abelian (for any fixed positive integer n). Then $e = e^n$ lies in G_n, the product $a^n b^n = (ab)^n$ is again an nth power, and $(a^n)^{-1} = (a^{-1})^n$ is again an nth power. Thus $G_n \leq G$. The hypothesis is sufficient rather than necessary; when $n = 1$, for example, $G_n = G$ for every group.

Exercise 5.57: No proper nontrivial subgroup forces cyclicity

If $G = \{e\}$ it is cyclic. Otherwise choose $a \neq e$. Its cyclic subgroup $\langle a \rangle$ is nontrivial because it contains a. By hypothesis it cannot be a proper subgroup, so $\langle a \rangle = G$. Thus G is cyclic. Choosing a nonidentity element is the step that excludes the trivial subgroup from this argument.

6 Cyclic Groups

Exercise 6.1: Quotient and nonnegative remainder

The division algorithm requires $n = mq + r$ with $0 \leq r < m$. Here

$$42 = 9(4) + 6, \qquad 0 \leq 6 < 9.$$

Thus the quotient is 4 and the remainder is 6. The nonnegative-remainder condition determines the quotient even when the dividend is negative.

Exercise 6.2: Quotient and nonnegative remainder

The division algorithm requires $n = mq + r$ with $0 \leq r < m$. Here

$$-42 = 9(-5) + 3, \qquad 0 \leq 3 < 9.$$

Thus the quotient is -5 and the remainder is 3. The nonnegative-remainder condition determines the quotient even when the dividend is negative.

Exercise 6.3: Quotient and nonnegative remainder

The division algorithm requires $n = mq + r$ with $0 \leq r < m$. Here

$$-50 = 8(-7) + 6, \qquad 0 \leq 6 < 8.$$

Thus the quotient is -7 and the remainder is 6. The nonnegative-remainder condition determines the quotient even when the dividend is negative.

Exercise 6.4: Quotient and nonnegative remainder

The division algorithm requires $n = mq + r$ with $0 \leq r < m$. Here

$$50 = 8(6) + 2, \qquad 0 \leq 2 < 8.$$

Thus the quotient is 6 and the remainder is 2. The nonnegative-remainder condition determines the quotient even when the dividend is negative.

Exercise 6.5: Greatest common divisor: Thirty-two and twenty-four

$32 = 1 \cdot 24 + 8$ and $24 = 3 \cdot 8 + 0$, so $\gcd(32, 24) = 8$. At each Euclidean step, the common divisors of the two current numbers are exactly those of the divisor and remainder; hence the final nonzero remainder is the greatest common divisor.

Exercise 6.6: Greatest common divisor: Forty-eight and eighty-eight

$88 = 1 \cdot 48 + 40$, $48 = 1 \cdot 40 + 8$, and $40 = 5 \cdot 8 + 0$. Thus $\gcd(48, 88) = 8$. At each Euclidean step, the common divisors of the two current numbers are exactly those of the divisor and remainder; hence the final nonzero remainder is the greatest common divisor.

Exercise 6.7: Greatest common divisor: Three hundred sixty and four hundred twenty

$420 = 1 \cdot 360 + 60$ and $360 = 6 \cdot 60 + 0$, so $\gcd(360, 420) = 60$. At each Euclidean step, the common divisors of the two current numbers are exactly those of the divisor and remainder; hence the final nonzero remainder is the greatest common divisor.

Exercise 6.8: Generators at order 5

If a generates a cyclic group of order 5, then a^k is a generator exactly when $\gcd(k, 5) = 1$. Indeed its order is $5/\gcd(k, 5)$. The eligible residues are $\{1, 2, 3, 4\}$, so the number of generators is $\varphi(5) = 4$.

Exercise 6.9: Generators at order 8

If a generates a cyclic group of order 8, then a^k is a generator exactly when $\gcd(k, 8) = 1$. Indeed its order is $8/\gcd(k, 8)$. The eligible residues are $\{1, 3, 5, 7\}$, so the number of generators is $\varphi(8) = 4$.

Exercise 6.10: Generators at order 12

If a generates a cyclic group of order 12, then a^k is a generator exactly when $\gcd(k, 12) = 1$. Indeed its order is $12/\gcd(k, 12)$. The eligible residues are $\{1, 5, 7, 11\}$, so the number of generators is $\varphi(12) = 4$.

Exercise 6.11: Generators at order 60

If a generates a cyclic group of order 60, then a^k is a generator exactly when $\gcd(k, 60) = 1$. Indeed its order is $60/\gcd(k, 60)$. The eligible residues are $\{1, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 49, 53, 59\}$, so the number of generators is $\varphi(60) = 16$.

Exercise 6.12: Automorphisms of Z 2

An additive map from $\mathbb{Z}_2$ is determined by the image k of 1: it must send x to $kx \pmod 2$. It is an automorphism exactly when k generates $\mathbb{Z}_2$, equivalently $\gcd(k, 2) = 1$. The choices are $k \in \{1\}$, so there are 1 automorphisms.

Exercise 6.13: Automorphisms of Z 6

An additive map from $\mathbb{Z}_6$ is determined by the image k of 1: it must send x to $kx \pmod 6$. It is an automorphism exactly when k generates $\mathbb{Z}_6$, equivalently $\gcd(k, 6) = 1$. The choices are $k \in \{1, 5\}$, so there are 2 automorphisms.

Exercise 6.14: Automorphisms of Z 8

An additive map from $\mathbb{Z}_8$ is determined by the image k of 1: it must send x to $kx \pmod 8$. It is an automorphism exactly when k generates $\mathbb{Z}_8$, equivalently $\gcd(k, 8) = 1$. The choices are $k \in \{1, 3, 5, 7\}$, so there are 4 automorphisms.

Exercise 6.15: Automorphisms of the integers

An additive map is determined by $\phi(1)$, since $\phi(n) = n\phi(1)$ for every integer, including negative integers. To be onto, this image must generate $\mathbb{Z}$, so it is 1 or -1. Both maps $n \mapsto n$ and $n \mapsto -n$ are automorphisms. Hence there are exactly 2.

Exercise 6.16: Automorphisms of Z twelve

The image of 1 can be any generator, and determines the whole map by $x \mapsto kx \pmod{12}$. The generators are $1, 5, 7, 11$, the residues relatively prime to 12. Each gives an invertible map (indeed each squares to 1 modulo 12), so there are 4 automorphisms.

Exercise 6.17: Twenty-five modulo thirty

The order of a residue k in the additive group $\mathbb{Z}_n$ is $n/\gcd(n, k)$. Here it is $30/5 = 6$. Directly, the generated subgroup is $\{0, 25, 20, 15, 10, 5\}$; the next multiple is zero and no previous positive multiple is zero.

Exercise 6.18: Thirty modulo forty-two

The subgroup has order $42/\gcd(42, 30) = 42/6 = 7$. Its elements, listed in increasing order, are $\{0, 6, 12, 18, 24, 30, 36\}$. They arise because integer multiples of 30 modulo 42 are exactly the multiples of 6 modulo 42.

Exercise 6.19: The powers of i

The powers are $1, i, -1, -i$, then 1 again. Thus the subgroup has exactly 4 elements. In particular $i^4 = 1$ and no smaller positive power equals 1.

Exercise 6.20: A normalized complex number

The generator $(1 + i)/\sqrt{2} = e^{i\pi/4}$ lies on the unit circle. Its kth power is 1 exactly when $k\pi/4 \in 2\pi\mathbb{Z}$, or $8 \mid k$. Hence its subgroup has 8 elements.

Exercise 6.21: The powers of one plus i

The modulus of $1 + i$ is $\sqrt{2}$. If $(1 + i)^m = (1 + i)^n$, taking moduli gives $(\sqrt{2})^m = (\sqrt{2})^n$, which forces $m = n$. Thus all integral powers are distinct and the generated subgroup is infinite (countably infinite). The modulus need not be 1 in the ambient group $\mathbb{C}^*$.

Exercise 6.22: All subgroups of Z 12

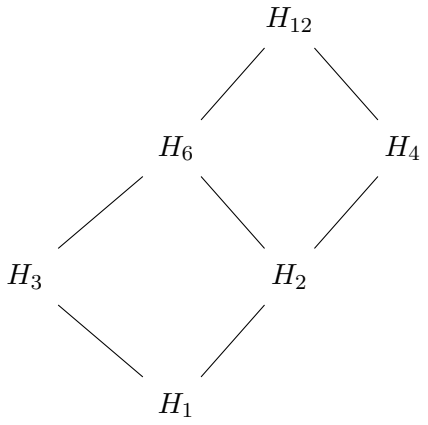
Every subgroup of a cyclic group is cyclic, and there is exactly one subgroup of each order d dividing 12. Label it $H_d = \langle 12/d \rangle$. Its complete element description is

$$H_d = \{0, 12/d, 2(12/d), \ldots, (d-1)(12/d)\}, \qquad |H_d| = d.$$

The complete list is specified by the following table:

d	1	2	3	4	6	12
generator n/d	12	6	4	3	2	1

For $d = 1$, the generator 12 represents 0 modulo 12, so $H_1 = \{0\}$; $H_{12} = \mathbb{Z}_{12}$. Inclusion satisfies $H_d \subseteq H_e$ exactly when $d \mid e$. The diagram joins only immediate inclusions (larger groups above):



Exercise 6.23: All subgroups of Z 36

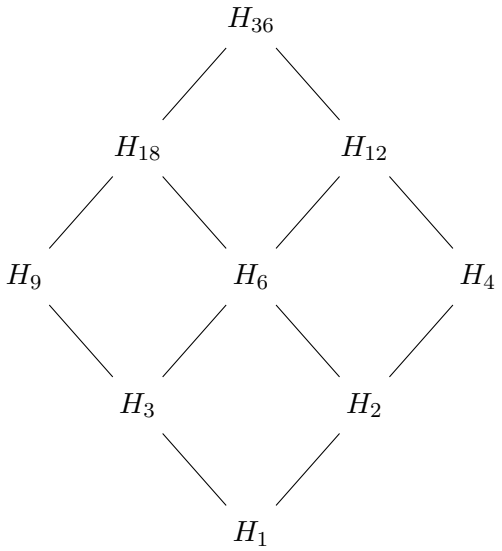
Every subgroup of a cyclic group is cyclic, and there is exactly one subgroup of each order d dividing 36. Label it $H_d = \langle 36/d \rangle$. Its complete element description is

$$H_d = \{0, 36/d, 2(36/d), \ldots, (d-1)(36/d)\}, \qquad |H_d| = d.$$

The complete list is specified by the following table:

d	1	2	3	4	6	9	12	18	36
generator n/d	36	18	12	9	6	4	3	2	1

For $d = 1$, the generator 36 represents 0 modulo 36, so $H_1 = \{0\}$; $H_{36} = \mathbb{Z}_{36}$. Inclusion satisfies $H_d \subseteq H_e$ exactly when $d \mid e$. The diagram joins only immediate inclusions (larger groups above):



Exercise 6.24: All subgroups of Z 8

Every subgroup of a cyclic group is cyclic, and there is exactly one subgroup of each order d dividing 8. Label it $H_d = \langle 8/d \rangle$. Its complete element description is

$$H_d = \{0, 8/d, 2(8/d), \ldots, (d-1)(8/d)\}, \qquad |H_d| = d.$$

The complete list is specified by the following table:

d	1	2	4	8
generator n/d	8	4	2	1

For $d = 1$, the generator 8 represents 0 modulo 8, so $H_1 = \{0\}$; $H_8 = \mathbb{Z}_8$. Inclusion satisfies $H_d \subseteq H_e$ exactly when $d \mid e$. The diagram joins only immediate inclusions (larger groups above):



Exercise 6.25: Subgroup orders in Z 6

The possible orders are exactly the positive divisors of 6:

$$\{1, 2, 3, 6\}.$$

A subgroup of order d actually exists for each listed divisor: it is generated by the residue $6/d$. Conversely the classification of subgroups of a cyclic group excludes all other orders.

Exercise 6.26: Subgroup orders in Z 8

The possible orders are exactly the positive divisors of 8:

$$\{1, 2, 4, 8\}.$$

A subgroup of order d actually exists for each listed divisor: it is generated by the residue $8/d$. Conversely the classification of subgroups of a cyclic group excludes all other orders.

Exercise 6.27: Subgroup orders in Z 12

The possible orders are exactly the positive divisors of 12:

$$\{1, 2, 3, 4, 6, 12\}.$$

A subgroup of order d actually exists for each listed divisor: it is generated by the residue $12/d$. Conversely the classification of subgroups of a cyclic group excludes all other orders.

Exercise 6.28: Subgroup orders in Z 20

The possible orders are exactly the positive divisors of 20:

$$\{1, 2, 4, 5, 10, 20\}.$$

A subgroup of order d actually exists for each listed divisor: it is generated by the residue $20/d$. Conversely the classification of subgroups of a cyclic group excludes all other orders.

Exercise 6.29: Subgroup orders in Z 17

The possible orders are exactly the positive divisors of 17:

$$\{1, 17\}.$$

A subgroup of order d actually exists for each listed divisor: it is generated by the residue $17/d$. Conversely the classification of subgroups of a cyclic group excludes all other orders.

Exercise 6.30: The definition of element order

An element a has order $n \in \mathbb{Z}^+$ if and only if n is the *least positive* integer such that $a^n = e$. Merely requiring $a^n = e$ is not sufficient: if a has order 2, it also satisfies $a^4 = e$, but its order is not 4. If no such positive exponent exists, the order is infinite.

Exercise 6.31: The definition of greatest common divisor

The definition is correct: the greatest common divisor of two positive integers is their largest positive common divisor. Existence is assured because 1 is a common divisor and all positive common divisors are bounded above by the smaller integer. A useful stronger property, proved via the Euclidean algorithm, is that every common divisor divides this greatest common divisor.

Exercise 6.32: Ten statements about cyclic groups

(a) True: powers of a common generator commute.

(b) False: the Klein four-group is abelian but not cyclic.

(c) False: if $q \neq 0$ generated $\mathbb{Q}$ additively, $q/2$ would not be an integer multiple of q.

(d) False: zero in a nontrivial additive cyclic group generates only its trivial subgroup.

(e) True: $\mathbb{Z}_n$ is abelian of order n for every positive n.

(f) False: the Klein four-group has order 4 and is not cyclic.

(g) False: 1 and 9 both generate $\mathbb{Z}_{20}$ but are not prime numbers; relative primality to 20 is the criterion.

(h) False in general: two groups may even have disjoint underlying sets, so their intersection is empty and cannot be a group. An operation shared in an ambient group is not specified here.

(i) True: the intersection contains the shared identity and is closed under the shared operation and inverses.

(j) True: if a generates a cyclic group of order greater than 2, then a^{-1} is a distinct generator; equality would imply $a^2 = e$ and order at most 2. The same reasoning covers infinite order.

Exercise 6.33: A finite noncyclic group

Use the Klein four-group $V = \{e, a, b, c\}$. Each nonidentity element has order 2, so every cyclic subgroup has at most two elements. None can equal the four-element group, proving that it is not cyclic.

Exercise 6.34: An infinite noncyclic group

The additive group $\mathbb{Q}$ is infinite and not cyclic. The zero element cannot generate it. If a nonzero rational q were a generator, then $q/2$ would have to be nq for some integer n; dividing by q gives $n = 1/2$, impossible.

Exercise 6.35: A cyclic group with one generator

The group $\mathbb{Z}_2$ works: the element 1 generates both residues, whereas 0 generates just itself. Hence it has exactly one generator. The trivial group is another possible example, since its identity generates it.

Exercise 6.36: An infinite cyclic group with four generators

No such group exists. Every infinite cyclic group is isomorphic to $\mathbb{Z}$. The only generators of $\mathbb{Z}$ are $1, -1$, since $k\mathbb{Z} = \mathbb{Z}$ forces $|k| = 1$. Isomorphisms preserve generators, so every infinite cyclic group has exactly two.

Exercise 6.37: A finite cyclic group with four generators

Use $\mathbb{Z}_5$. Its generators are $1, 2, 3, 4$, since each is relatively prime to 5 and therefore has order 5. The identity 0 is not a generator, so there are exactly four.

Exercise 6.38: Primitive roots for n equals 4

The roots of unity are $e^{2\pi ik/4}$, and such a root is primitive exactly when $\gcd(k,4)=1$, since its order is $4/\gcd(k,4)$. Therefore the primitive roots are

$$\{e^{2\pi ik/4} : k \in \{1,3\}\}.$$

These are $i, -i$.

Exercise 6.39: Primitive roots for n equals 6

The roots of unity are $e^{2\pi ik/6}$, and such a root is primitive exactly when $\gcd(k,6)=1$, since its order is $6/\gcd(k,6)$. Therefore the primitive roots are

$$\{e^{2\pi ik/6} : k \in \{1,5\}\}.$$

These are $1/2 + i\sqrt{3}/2$ and $1/2 - i\sqrt{3}/2$.

Exercise 6.40: Primitive roots for n equals 8

The roots of unity are $e^{2\pi ik/8}$, and such a root is primitive exactly when $\gcd(k,8)=1$, since its order is $8/\gcd(k,8)$. Therefore the primitive roots are

$$\{e^{2\pi ik/8} : k \in \{1,3,5,7\}\}.$$

These are $(1+i)/\sqrt{2}$, $(-1+i)/\sqrt{2}$, $(-1-i)/\sqrt{2}$, and $(1-i)/\sqrt{2}$.

Exercise 6.41: Primitive roots for n equals 12

The roots of unity are $e^{2\pi ik/12}$, and such a root is primitive exactly when $\gcd(k,12)=1$, since its order is $12/\gcd(k,12)$. Therefore the primitive roots are

$$\{e^{2\pi ik/12} : k \in \{1,5,7,11\}\}.$$

Their arguments are $\pi/6, 5\pi/6, 7\pi/6, 11\pi/6$; equivalently the roots are $(\pm\sqrt{3} \pm i)/2$ with independent signs.

Exercise 6.42: Synopsis of cyclic implies abelian

Express any two elements as powers of one generator and use addition of their integer exponents to see that their products in the two orders agree.

Exercise 6.43: Synopsis of the cyclic-subgroup theorem

For a nontrivial subgroup of a cyclic group, choose the least positive exponent whose generator power belongs to the subgroup. Divide the exponent of any other subgroup element by this least exponent; closure puts the remainder power in the subgroup, and minimality forces that remainder to be zero. Thus every element of the subgroup is a power of the chosen one.

Exercise 6.44: A homomorphism from a cyclic group is determined by one value

Write an arbitrary $x \in G$ as a^m for an integer m. Preservation of products and inverses gives $\phi(x) = \phi(a^m) = \phi(a)^m$. If $\psi(a) = \phi(a)$, then $\psi(x) = \psi(a)^m = \phi(a)^m = \phi(x)$. This works for all x, including the identity and negative powers. Thus the maps agree everywhere once they agree on a generator.

Exercise 6.45: Integer linear combinations form a subgroup

Let $H = \{nr + ms : n, m \in \mathbb{Z}\}$. It contains $0 = 0r + 0s$. The sum of $nr + ms$ and $n'r + m's$ is $(n+n')r + (m+m')s$, again in H. The negative is $(-n)r + (-m)s$, also in H. Hence H is a subgroup of the additive group $\mathbb{Z}$ (indeed it is $\gcd(r, s)\mathbb{Z}$).

Exercise 6.46: The orders of ab and ba

The elements are conjugate: $ba = a^{-1}(ab)a$. Therefore for every positive integer k,

$$(ba)^k = a^{-1}(ab)^k a,$$

proved by cancellation of the adjacent aa^{-1} factors. Consequently $(ba)^k = e$ if and only if $(ab)^k = e$. Their least positive identity exponents are therefore the same, so ba also has order n. This proves equality, rather than merely showing its order divides n.

Exercise 6.47: Least common multiples via subgroups

(a) The intersection $r\mathbb{Z} \cap s\mathbb{Z}$ is a subgroup of $\mathbb{Z}$ containing $rs > 0$. Its unique positive generator is $\ell = \operatorname{lcm}(r, s)$: its positive members are exactly the positive common multiples, and its least positive member generates it.

(b) $\ell = rs$ exactly when $\gcd(r, s) = 1$.

(c) Put $d = \gcd(r, s)$ and write $r = du, s = dv$ with $\gcd(u, v) = 1$. If M is a common multiple, write $M = duk$. Since $dv \mid M$, we have $v \mid uk$; relatively prime u, v imply $v \mid k$ (multiply a Bezout identity for u, v by k). Thus $duv \mid M$. Conversely duv is a common multiple. Therefore $\ell = duv$, and $d\ell = d^2uv = rs$. Since r, s are positive, this formula also proves the equivalence in part (b).

Exercise 6.48: Finitely many subgroups force a finite group

First, no element can have infinite order. If a did, its infinite cyclic subgroup would contain the infinitely many distinct subgroups $\langle a^m \rangle$, $m = 1, 2, \ldots$, all also subgroups of G. They are distinct because in an infinite cyclic group the exponent sets $m\mathbb{Z}$ are distinct. Thus every cyclic subgroup of G is finite. There are only finitely many subgroups by hypothesis, hence only finitely many cyclic subgroups. Every element lies in the cyclic subgroup it generates, so G is a finite union of finite cyclic subgroups and is finite.

Exercise 6.49: All proper subgroups cyclic does not suffice

The Klein four-group is not cyclic, since every element has order at most 2. Its proper subgroups are $\{e\}$, $\{e, a\}$, $\{e, b\}$, and $\{e, c\}$, all cyclic. Indeed any subgroup containing two different nonidentity elements must also contain their product, the third nonidentity element, and hence must be all of V. Thus this is the required counterexample.

Exercise 6.50: A unique element of order two is central

For arbitrary $x \in G$, compute $(xax^{-1})^2 = xa^2x^{-1} = e$. Also $xax^{-1} \neq e$, because otherwise multiplying by x^{-1} on the left and x on the right would give $a = e$. Therefore this conjugate also has order 2, and uniqueness forces $xax^{-1} = a$. Multiplying on the right by x yields $xa = ax$. Since x was arbitrary, a commutes with every element.

Exercise 6.51: Generators at order pq

A residue modulo pq generates the cyclic group exactly when it is not divisible by p or by q. Among the pq residues $0, \ldots, pq-1$, there are q multiples of p, p multiples of q, and one residue (zero) counted in both lists. Inclusion-exclusion gives

$$pq - q - p + 1 = (p-1)(q-1)$$

generators. Distinctness and primality of p, q ensure that these are exactly the forbidden prime divisors.

Exercise 6.52: Generators at a prime-power order

A residue modulo p^r generates the additive group precisely when it is relatively prime to p^r, equivalently not divisible by p. Among $0, 1, \ldots, p^r - 1$, the multiples of p are $0, p, 2p, \ldots, (p^{r-1} - 1)p$, exactly p^{r-1} residues. Therefore the number of generators is

$$p^r - p^{r-1} = p^{r-1}(p-1).$$

This includes $r = 1$, giving $p - 1$ nonzero generators in $\mathbb{Z}_p$.

Exercise 6.53: A power equation when the exponent divides the order

Let $G = \langle a \rangle$ have order n, and write $n = mq$. Every element has a unique representation a^k with $0 \leq k < n$. The equation $(a^k)^m = e$ holds if and only if $n \mid km$, which, since $n = mq$, is equivalent to $q \mid k$. The eligible exponents are therefore

$$0, q, 2q, \ldots, (m-1)q,$$

exactly m distinct residues. The corresponding m distinct group elements are all the solutions. The argument includes the cases $m = 1$ and $m = n$.

Exercise 6.54: A power equation for a general exponent

Put $d = \gcd(n, m)$ and write $n = dn'$, $m = dm'$ with $\gcd(n', m') = 1$. For $x = a^k$, the equation $x^m = e$ is equivalent to $n \mid mk$, hence to $n' \mid m'k$. Relative primality makes this equivalent to $n' \mid k$. There are exactly d such residues modulo n, namely $k = 0, n', \ldots, (d-1)n'$. Thus there are $\gcd(n, m)$ solutions. In the question's case $1 < m < n$ and $m \nmid n$, this number is strictly less than m (and equals 1 if n, m are relatively prime).

Exercise 6.55: Subgroups at prime order

Every subgroup of $\mathbb{Z}_p$ has order dividing p, by the classification of subgroups of a cyclic group. Since p is prime, the only divisors are 1 and p, yielding respectively the trivial subgroup and the whole group. Hence there is no proper nontrivial subgroup. Equivalently, any nonzero residue k is relatively prime to p and itself generates all of $\mathbb{Z}_p$.

Exercise 6.56: Finding a cyclic subgroup of a least-common-multiple order

Write $H = \langle a \rangle$, $K = \langle b \rangle$, with a, b of orders r, s.

(a) Assume $\gcd(r, s) = 1$. Any element of $H \cap K$ has order dividing both r and s, hence is the identity. Since the group is abelian, $(ab)^{rs} = e$. Conversely if $(ab)^t = e$, then $a^t = b^{-t}$ lies in $H \cap K$, so $a^t = b^t = e$. Thus $r \mid t$ and $s \mid t$, giving $rs \mid t$. Therefore ab has order exactly rs, and $\langle ab \rangle$ is the required subgroup.

(b) Factor r, s into prime powers. For each prime p appearing, take the larger of its two prime-power factors, say p^{γ_p}. If that factor comes from r, choose $c_p = a^{r/p^{\gamma_p}}$; its order is exactly p^{γ_p}, by the order formula for a power. If it comes from s, choose $c_p = b^{s/p^{\gamma_p}}$ instead (ties can use either). All these chosen elements commute because G is abelian. Their orders are pairwise relatively prime, so applying part (a) repeatedly shows that the product $c = \prod_p c_p$ has order $\prod_p p^{\gamma_p} = \operatorname{lcm}(r, s)$. Thus $\langle c \rangle$ is the required cyclic subgroup. If $r = s = 1$, simply take $c = e$.

The product ab itself need not have the least-common-multiple order when r, s overlap; the prime-power selection prevents cancellation of common factors.

7 Generating Sets and Cayley Digraphs

Exercise 7.1: Two and three modulo twelve

The generated subgroup contains $3-2=1$, because subgroups are closed under addition and negatives. Once it contains 1, it contains every residue. Therefore it is $\mathbb{Z}_{12}=\{0,1,2,3,4,5,6,7,8,9,10,11\}$.

Exercise 7.2: Four and six modulo twelve

Every integer combination of 4 and 6 is even, so the subgroup lies inside the even residues. It contains $6-4=2$, and hence every even residue. Thus it is exactly $\{0,2,4,6,8,10\}$.

Exercise 7.3: Eight and ten modulo eighteen

Every combination of $8,10$ and the modulus 18 is even, so no odd residue can occur. The difference $10-8=2$ belongs, and its multiples give all even residues. Thus the subgroup is $\{0,2,4,6,8,10,12,14,16\}$.

Exercise 7.4: Twelve and thirty modulo thirty-six

The difference $30-2\cdot 12=6$ belongs. Every input and the modulus are divisible by 6, so only multiples of 6 occur, and all of them occur by adding 6. The subgroup is therefore $\{0,6,12,18,24,30\}$.

Exercise 7.5: Twelve and forty-two in the integers

All generated elements are combinations $12m+42n$, hence multiples of 6. Conversely $42-3\cdot 12=6$ belongs, so every integer multiple of 6 belongs. The subgroup is $6\mathbb{Z}=\{\ldots,-12,-6,0,6,12,\ldots\}$.

Exercise 7.6: Three integer generators

All three inputs $18,24,39$ are divisible by 3. The generated subgroup contains $24-18=6$ and then $39-6\cdot 6=3$. It therefore contains every multiple of 3, and contains no other integer. The answer is $3\mathbb{Z}=\{\ldots,-6,-3,0,3,6,\ldots\}$.

Exercise 7.7: Products in the eight-element digraph

The outer solid cycle gives $a^4=e$. Dashed edges give $b^2=e$, and the reversed inner solid cycle gives $ba=a^{-1}b$. Thus $ba^k=a^{-k}b$ for every integer k.

(a) $(a^2b)a^3=a^2a^{-3}b=a^{-1}b=a^3b$.

(b) $(ab)(a^3b)=aa^{-3}b^2=a^{-2}=a^2$.

(c) $b(a^2b)=a^{-2}b^2=a^{-2}=a^2$.

Each result can also be checked by starting at the first factor's vertex and following the right factor's directed edge sequence.

Exercise 7.8: The square digraph table

Both indicated generator types are involutions, and traversing one of each takes e to c in either order. Thus $a^2 = b^2 = e$, $ab = ba = c$, giving

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

For an order-two generator an undirected edge stands for the two opposite directed edges. The resulting group is the Klein four-group.

Exercise 7.9: The alternating-edge hexagon table

Read the two edge types as involutions a, b. The paths give $c = ab$, $d = ba$, and $f = aba = bab$. One concrete model is $a = (12)$ and $b = (23)$, so $c = (123)$, $d = (132)$, and $f = (13)$. Multiplying these permutations, with the rightmost one applied first, yields the table in the requested alphabetical order:

	e	a	b	c	d	f
e	e	a	b	c	d	f
a	a	e	c	b	f	d
b	b	d	e	f	a	c
c	c	f	a	d	e	b
d	d	b	f	e	c	a
f	f	c	d	a	b	e

For example $ab = c$ but $ba = d$, so this six-element group is nonabelian. The model also verifies the whole table's associativity.

Exercise 7.10: The two-triangle digraph table

The outside arrows give $a^3 = e$ and $c = a^2$. The dashed edges give $b^2 = e$, $ab = f$, $a^2b = d$. Reading the inside arrows yields $ba = a^{-1}b$, so this is the triangular dihedral group. Using $a = (123)$, $b = (12)$ realizes these labels concretely and gives

	e	a	b	c	d	f
e	e	a	b	c	d	f
a	a	c	f	e	b	d
b	b	d	e	f	a	c
c	c	e	d	a	f	b
d	d	f	c	b	e	a
f	f	b	a	d	c	e

For example $ab = f$ and $ba = d$, consistent with the distinct paths in the figure. This diagram and the preceding one describe isomorphic groups, but their element names represent different permutations.

Exercise 7.11: Recognizing commutativity from paths

Compare the endpoints of paths using two generator types in opposite orders. If some s-then-t path ends differently from the corresponding t-then-s path at the same starting point, the group is noncommutative. Conversely, if every pair of generators commutes, their inverses also commute; any two words in generators and inverses can then be reordered past each other. Since such words are all group elements, the group is abelian. Thus this two-step-path comparison is a complete criterion.

Exercise 7.12: The eight-element example is nonabelian

Starting at e, traversing an a edge and then a b edge ends at ab, whereas traversing b then a ends at a^3b. These are different labeled vertices in Fig. 7.11(b). Therefore $ab \neq ba$ and the group is noncommutative.

Exercise 7.13: Recognizing cyclicity

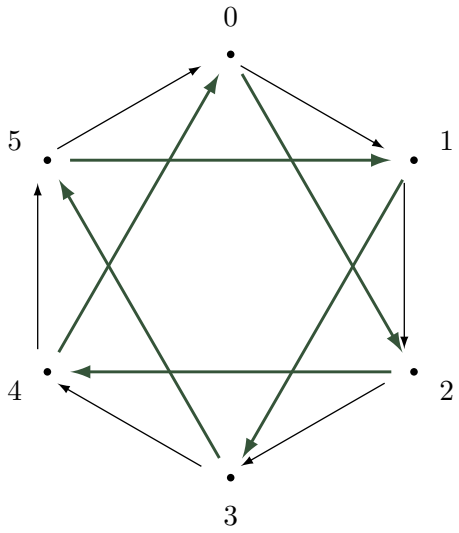
A single generator type forming a directed cycle through every vertex proves cyclicity, but its absence does not disprove cyclicity. A diagram might use a redundant generating set without displaying any individual generator of the whole group. For example $\mathbb{Z}_6$ generated by $\{2, 3\}$ is cyclic, while the 2-edges form triangles and the 3-edges form pairs. Thus cyclicity need not be visually apparent from one edge type; one may need a repeated word in several edge types.

Exercise 7.14: The inner triangle is not a subgroup

The smaller triangle in Fig. 7.9(b) has the vertices $\{1, 3, 5\}$. It does not contain the additive identity 0, so it cannot be a subgroup of $\mathbb{Z}_6$. It is also not closed: $1 + 1 = 2$ is outside it. It is a translated copy of the subgroup $\{0, 2, 4\}$, which explains the identical triangle shape without making it a subgroup.

Exercise 7.15: A Cayley digraph with generators one and two

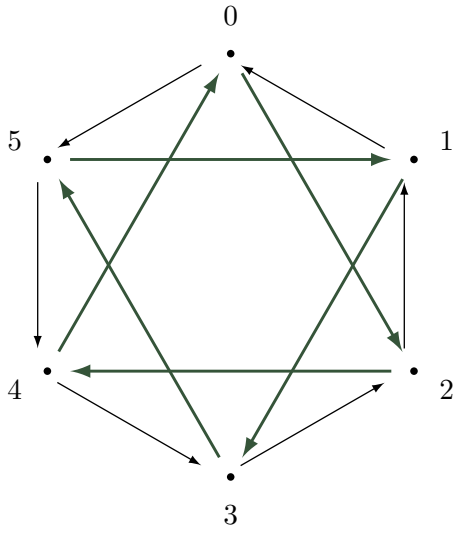
Place the six residues around a hexagon in the order $0, 1, 2, 3, 4, 5$ clockwise. Draw a thin black arrow $x \to x + 1$ and a thick green arrow $x \to x + 2$, reducing modulo 6:



The black arrows form $0 \to 1 \to 2 \to 3 \to 4 \to 5 \to 0$. The green arrows form $0 \to 2 \to 4 \to 0$ and $1 \to 3 \to 5 \to 1$. Thus each vertex has exactly one incoming and one outgoing arrow of each generator type, as required.

Exercise 7.16: A Cayley digraph with generators two and five

Use the same labeled hexagon. Draw thick green arrows $x \to x+2$ and thin black arrows $x \to x+5 = x - 1$ modulo 6:



The green triangles are $0 \to 2 \to 4 \to 0$ and $1 \to 3 \to 5 \to 1$. The black outside cycle is $0 \to 5 \to 4 \to 3 \to 2 \to 1 \to 0$. Since 5 alone is a generator, the combined diagram is connected and represents all of $\mathbb{Z}_6$.

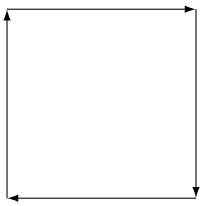
Exercise 7.17: Relations read from closed paths

(a) Begin at the identity and follow a closed path. Write down the generator for each forward edge and its inverse for each backward edge. The resulting word is a product equal to e, hence a relation. Conversely any such relation returns the identity to itself when its word is followed.

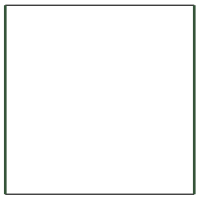
(b) In Fig. 7.11(b), the outside four-cycle gives $a^4 = e$, a dashed round trip gives $b^2 = e$, and the path expressing $ba = a^{-1}b$ gives $baba = e$. Thus three relations are $a^4 = e$, $b^2 = e$, and $(ba)^2 = e$. The last can also be written $bab = a^{-1}$.

Exercise 7.18: Minimal digraphs at order four

For the cyclic group use one generator: a directed four-cycle. For the Klein group one generator never suffices, since every element has order at most 2; use two involutions and a square with different edge types in the two directions. Here a double-headed edge is represented by an undirected line, as usual for an involution.



$\mathbb{Z}_4$: one generator



V: two generators

Each square has four vertices at its corners. In the right-hand diagram horizontal edges have one generator type and vertical edges the other; both types are their own reverses.

Exercise 7.19: Two involutions generating a nonabelian group of order twice n

Take symmetries of a regular n-gon, $n \geq 3$. Let r rotate through $2\pi/n$ and s reflect in an axis through its center. The symmetries are exactly r^k and $r^k s$ for $0 \leq k < n$: these $2n$ transformations are distinct because the first preserve orientation and the second reverse it, and the rotations themselves are distinct. They are closed under composition because $sr = r^{-1}s$.

Choose two order-two generators.

The elements s and rs are reflections. They satisfy $s^2 = e$ and $(rs)^2 = r(sr)s = rr^{-1}s^2 = e$, and neither is the identity. Their product $(rs)s = r$ gives the basic rotation, so they generate every $r^k, r^k s$.

Check noncommutativity.

The products rs and $sr = r^{-1}s$ differ: equality would give $r = r^{-1}$, contradicting that r has order $n \geq 3$. Thus the group is nonabelian, has exactly $2n$ elements, and is generated by two elements of order 2.